

PROJECT REPORT

NAME : N.SRI SOUNDARRAJ
COURSE : CLOUD COMPUTING

TASK -1

Windows Administration Server & Client OS.

TASK -2

Linux Administration & Commands.

TASK-1

Windows Server & Client :

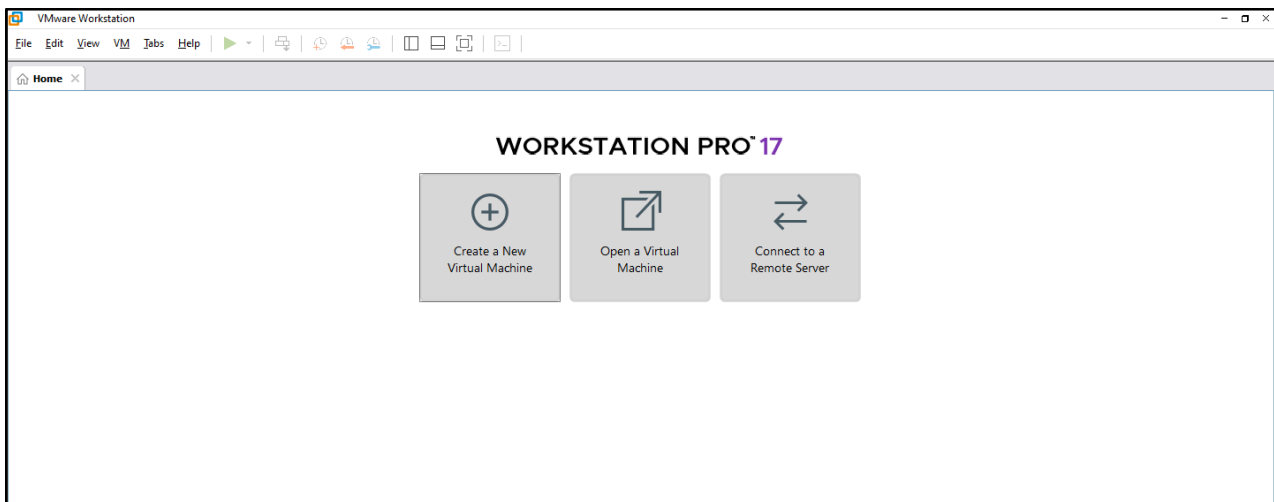
1.Prerequisites:

1. VMware product installed (Workstation Pro/Player on your desktop).
2. Windows Server ISO (evaluation can be downloaded from Microsoft EvalCenter).
3. Host resources (minimum for a lab VM):
 - 2 vCPU.
 - 4–8 GB RAM (8 GB recommended for GUI).
 - 40–60 GB virtual disk.

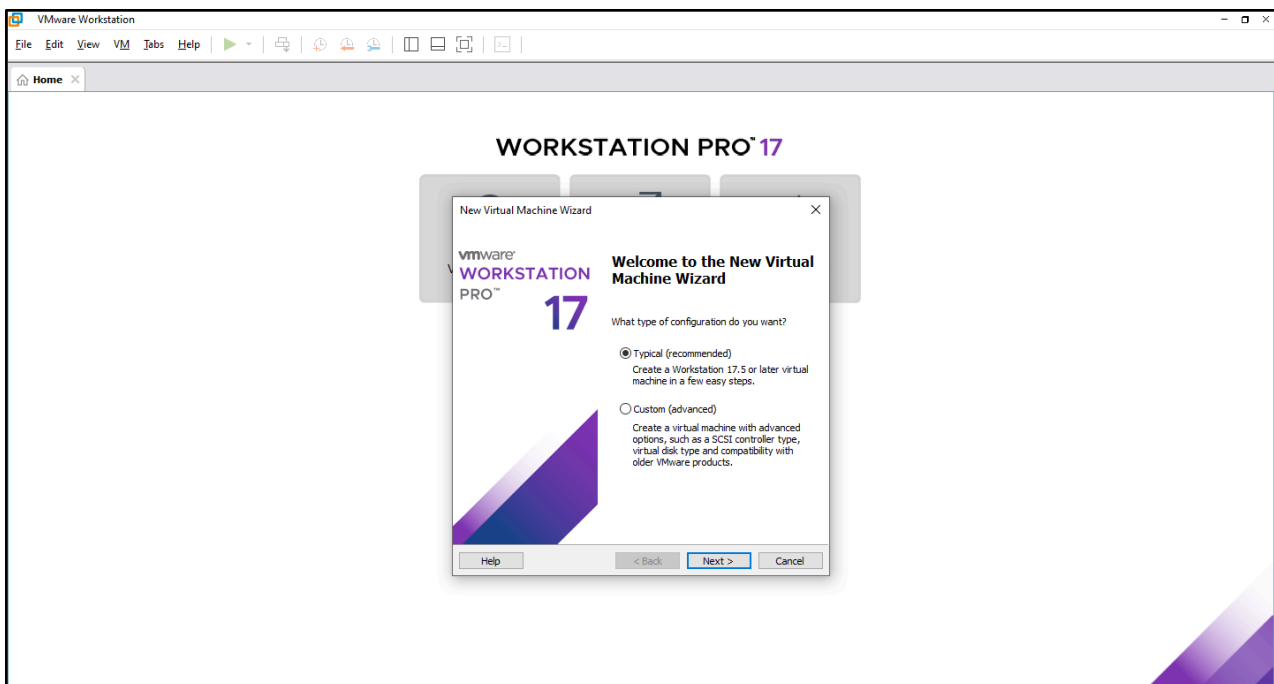
2.Create the virtual machine:

In VMware Workstation (GUI flow; ESXi wizard is almost identical).

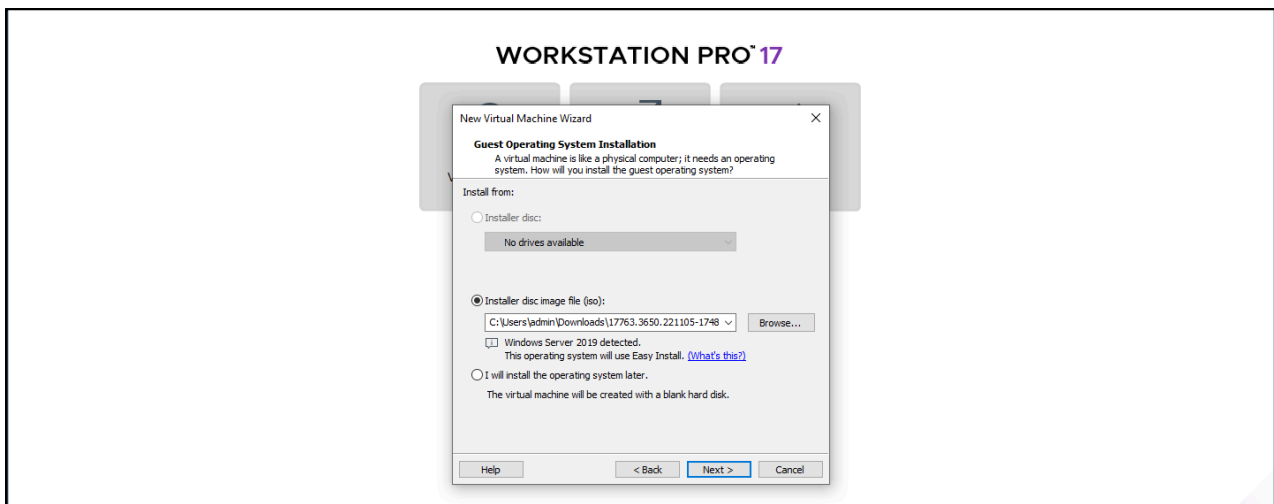
1.Open VMware → click Create a New Virtual Machine.



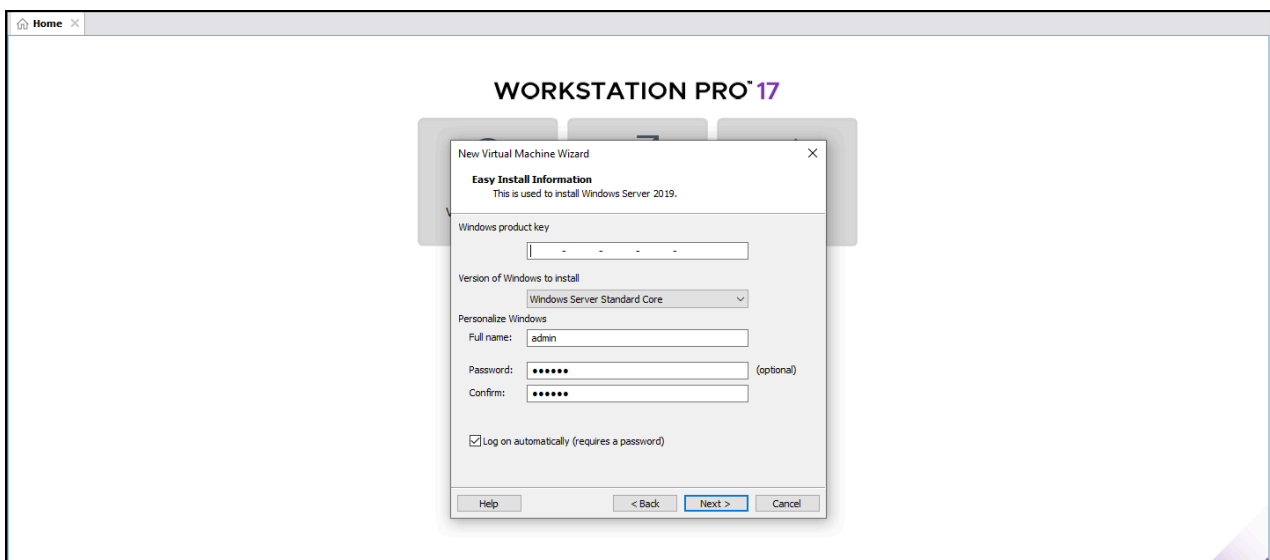
2. Choose Typical (or Custom if you want fine-grained control) → Next.



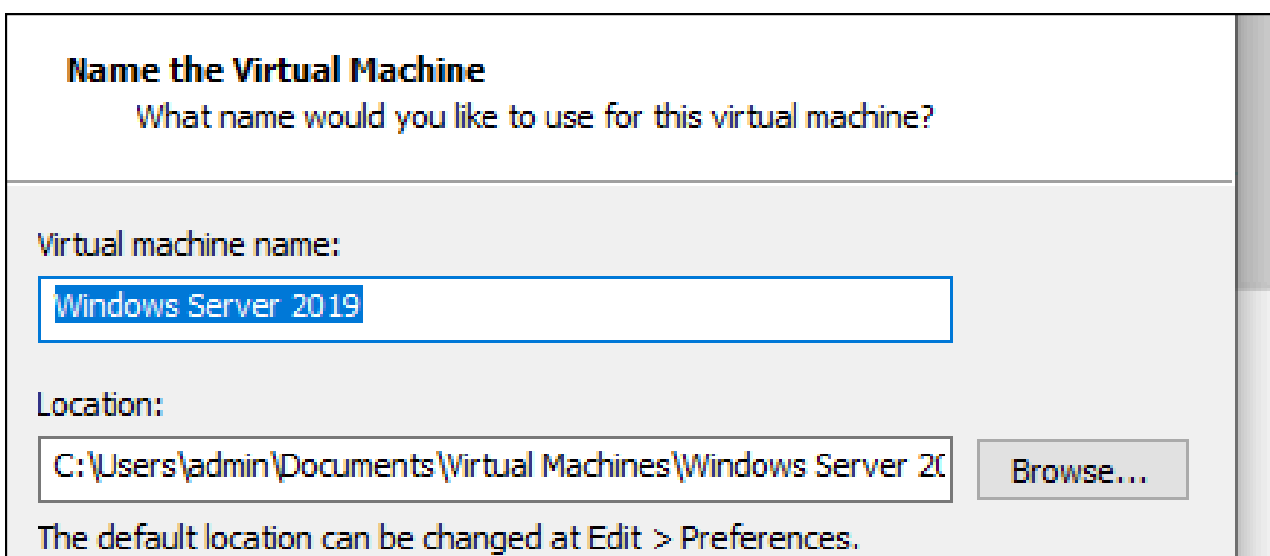
3. Select Installer disc image file (iso) and browse to your Windows Server ISO.
4. Set Guest OS to Microsoft Windows and Version to Windows Server.



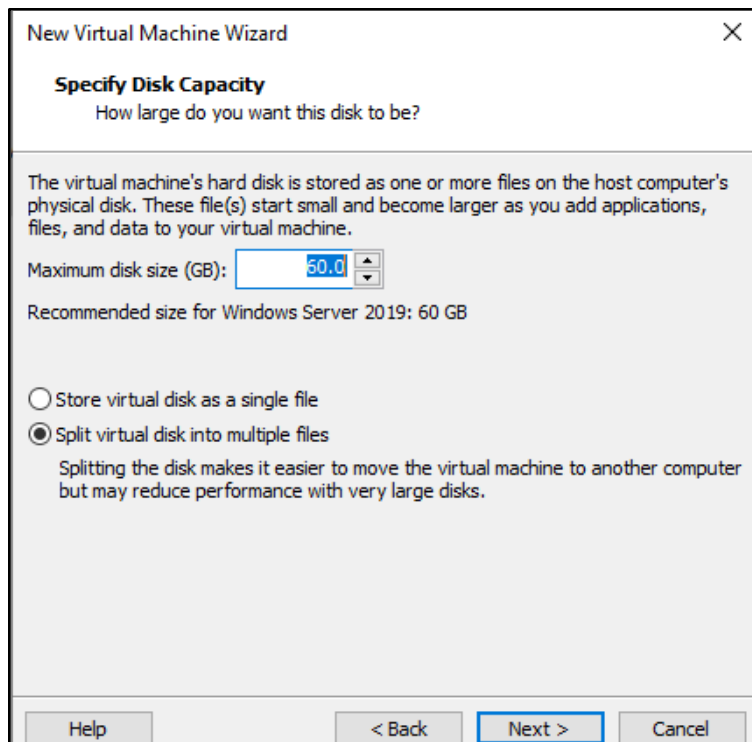
5. Select windows server standard core option and then set username,password(optional).



6. Name the VM and choose the VM location on disk.

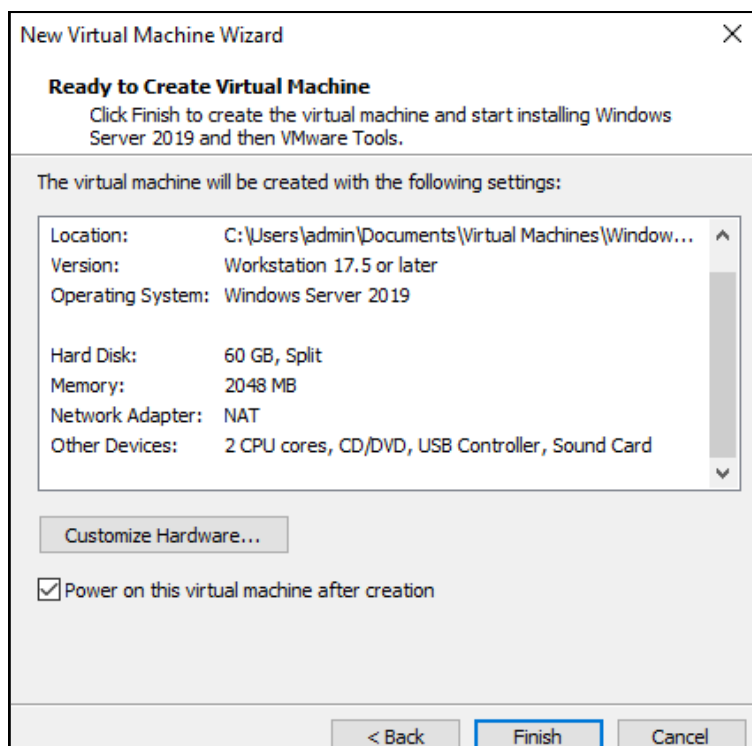


7. Allocate CPU: set Number of processors to 1 and Number of cores to 2 (or as your host allows).
8. Allocate memory: set at least 4096 MB, preferably 8192 MB for Desktop Experience.
9. Configure network: use NAT or Bridged depending on your lab design.
10. Create a virtual disk:
 - Size: at least 40–60 GB.
 - Store as a single file (slightly faster) or split.



The screenshot shows the 'Specify Disk Capacity' step of the 'New Virtual Machine Wizard'. The title bar reads 'New Virtual Machine Wizard'. The main heading is 'Specify Disk Capacity' with the subtitle 'How large do you want this disk to be?'. A text box explains: 'The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.' Below this, there is a 'Maximum disk size (GB):' label followed by a text input field containing '60.0' and a spinner control. A note states 'Recommended size for Windows Server 2019: 60 GB'. Two radio buttons are present: 'Store virtual disk as a single file' (unselected) and 'Split virtual disk into multiple files' (selected). A note below the selected option says: 'Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.' At the bottom are buttons for 'Help', '< Back', 'Next >', and 'Cancel'.

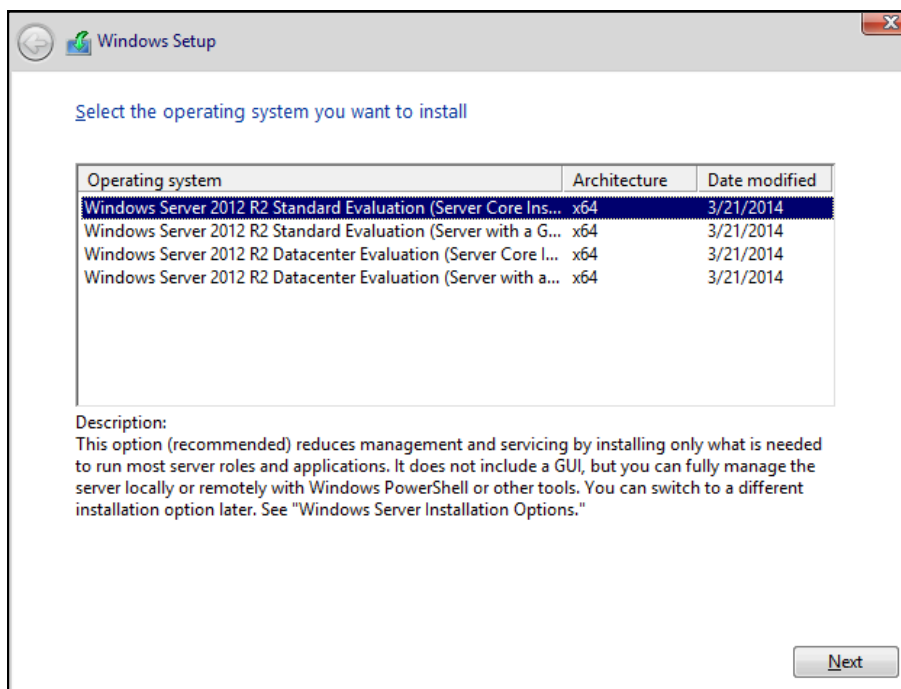
11. Finish the wizard; the empty VM will be created.



The screenshot shows the 'Ready to Create Virtual Machine' step of the 'New Virtual Machine Wizard'. The title bar reads 'New Virtual Machine Wizard'. The main heading is 'Ready to Create Virtual Machine' with the subtitle 'Click Finish to create the virtual machine and start installing Windows Server 2019 and then VMware Tools.' Below this, a section titled 'The virtual machine will be created with the following settings:' contains a list of settings: Location: C:\Users\admin\Documents\Virtual Machines\Window..., Version: Workstation 17.5 or later, Operating System: Windows Server 2019, Hard Disk: 60 GB, Split, Memory: 2048 MB, Network Adapter: NAT, and Other Devices: 2 CPU cores, CD/DVD, USB Controller, Sound Card. A 'Customize Hardware...' button is located below the settings list. A checkbox labeled 'Power on this virtual machine after creation' is checked. At the bottom are buttons for '< Back', 'Finish', and 'Cancel'.

3. Boot and install Windows Server :

1. Power on the VM; it should boot from the ISO.
2. When prompted, choose:
 - Language, Time, Keyboard → Next → Install now.
3. Click I don't have a product key (for evaluation), or enter a key if you have one.
4. Choose edition: typical lab choice is Windows Server Standard (Desktop Experience) or Datacenter (Desktop Experience) → Next.

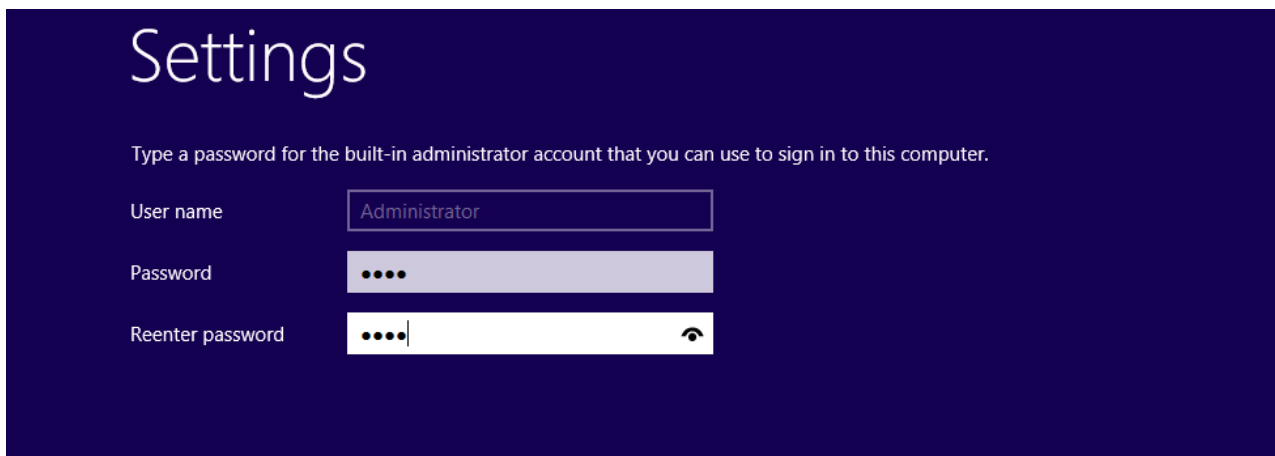


5. Choose Custom: Install Windows only (advanced).
6. Select the unallocated disk → New (optional) → Apply → Next to start installation.

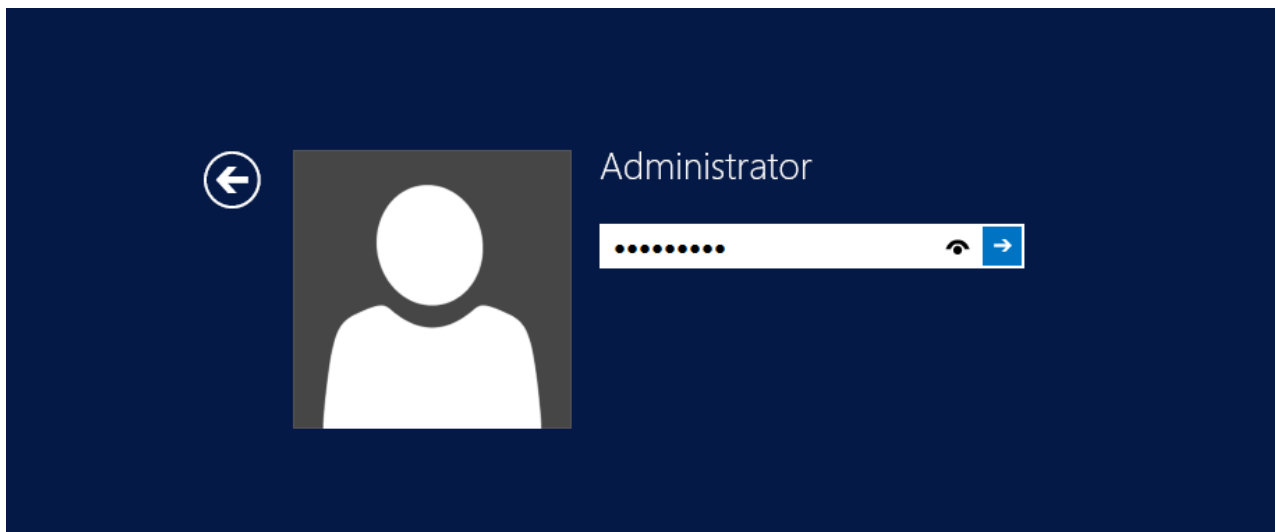


7.Wait for copying files, automatic reboots, then you will be prompted to set the local Administrator password.

8.Press Ctrl+Alt+Del (VMware menu option), log in as Administrator with the password you just set.



9.The windows Server boot on Administrator password UI.

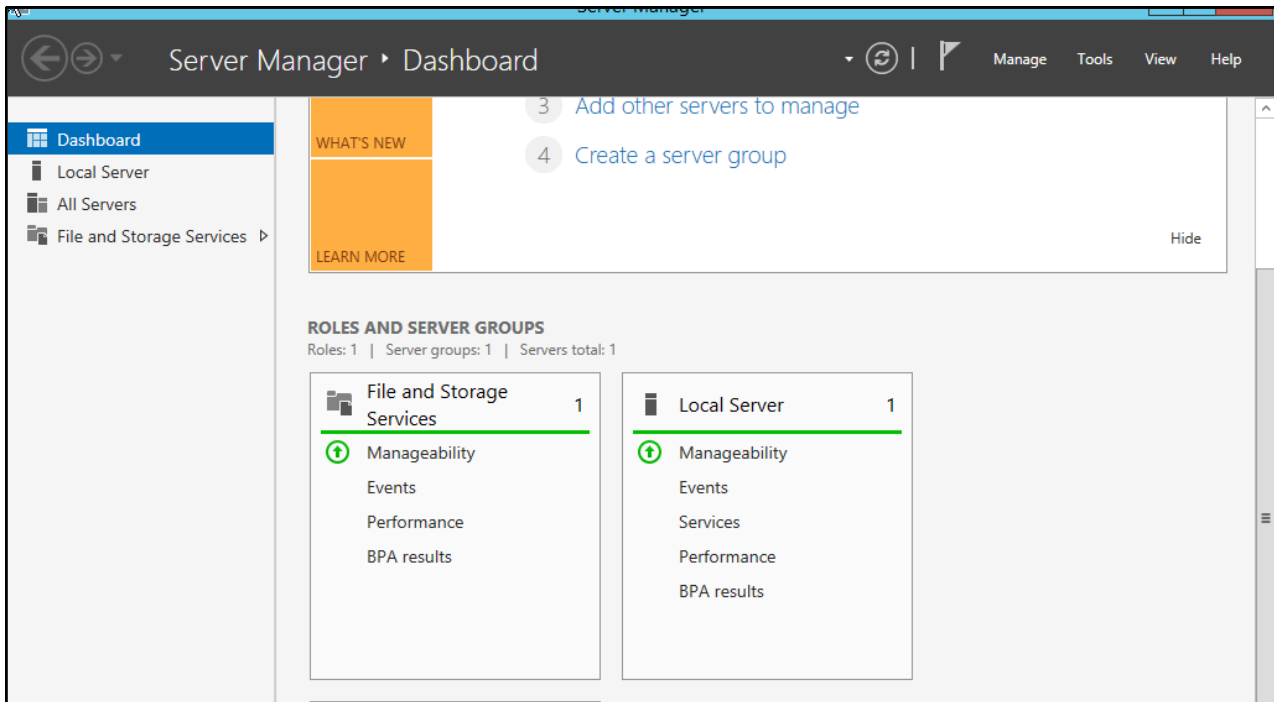


10.This is Windows Server home UI.

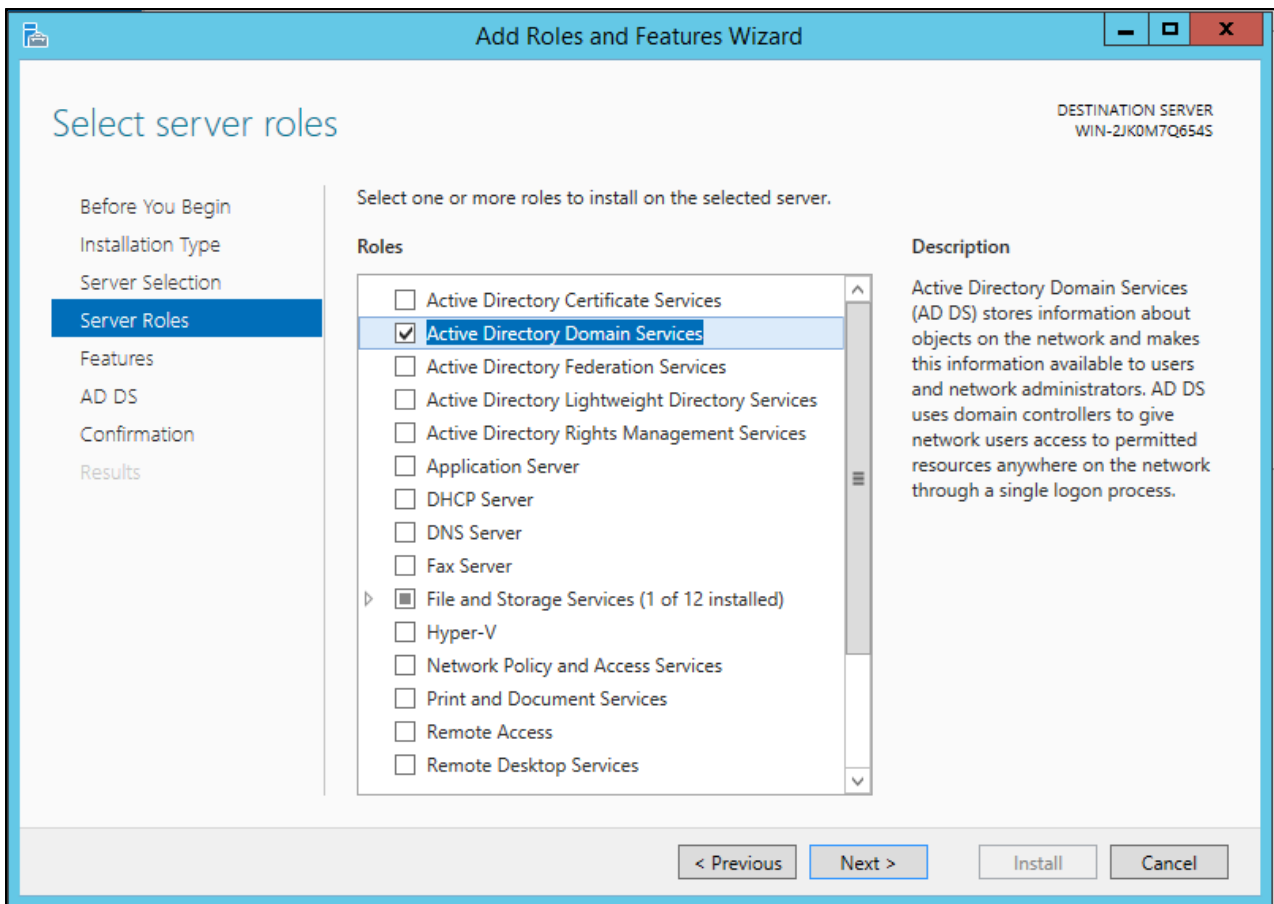


4. Install and Configure Active Directory Domain Service :

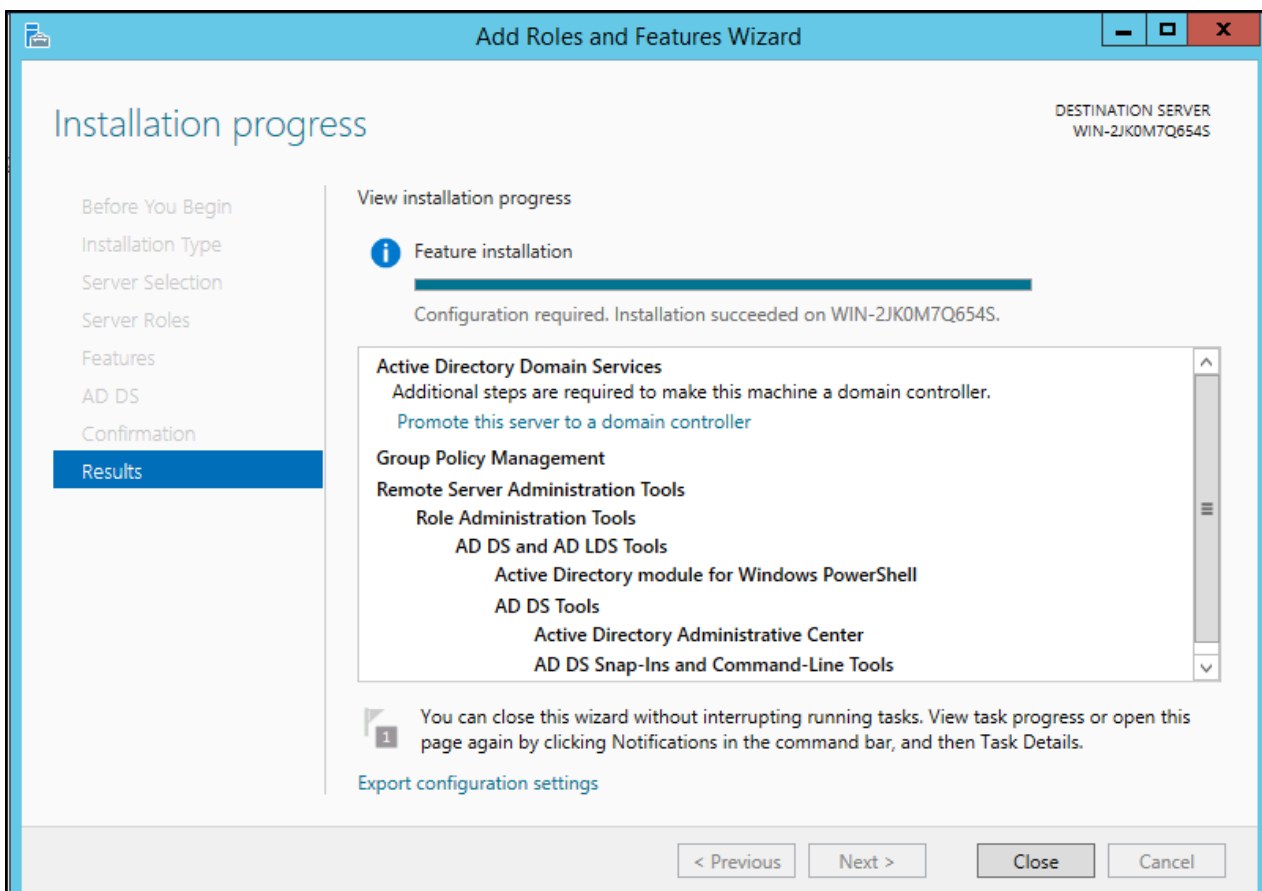
1. Open Server Manager → Add roles and features.
2. Role-based installation → select this server.



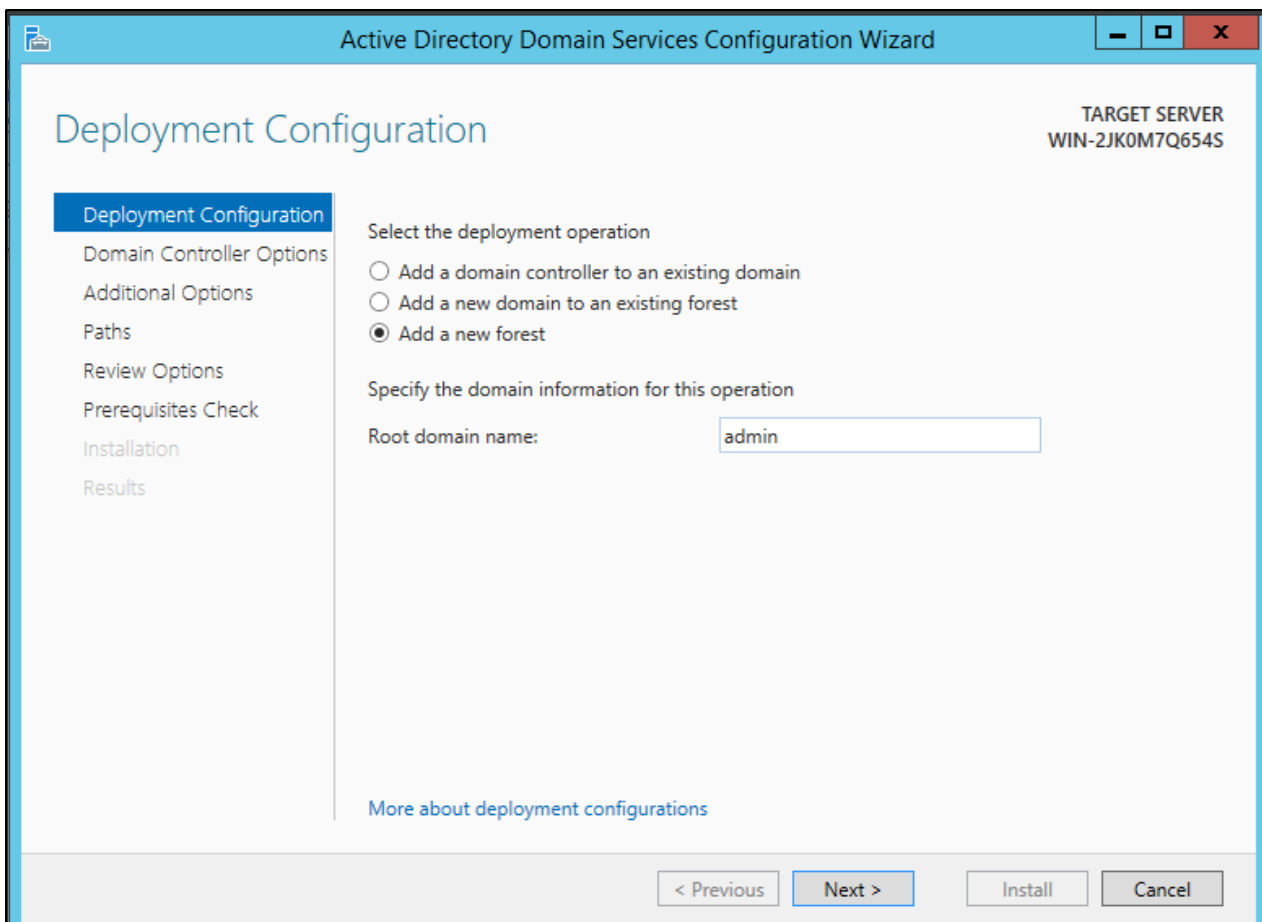
3. Tick Active Directory Domain Services; when prompted, also add required features (DNS will be added automatically).
4. Next → Install, then close when done.



5. In Server Manager, click the yellow flag → Promote this server to a domain controller.



6. Choose Add a new forest; root domain name sri.domain



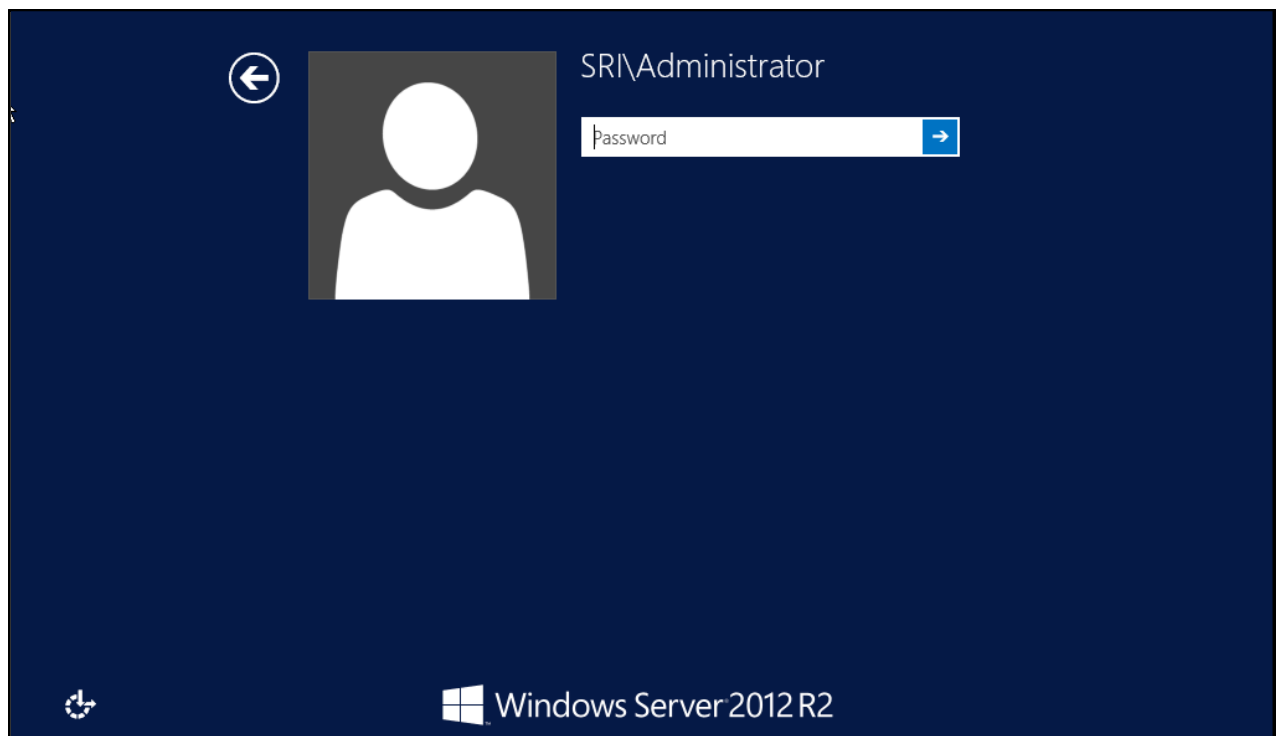
7. Set Directory Services Restore Mode (DSRM) password → Next through DNS/paths → Install; the server will reboot.

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes 'Server Manager ▸ Dashboard' and a 'Manage' button. The main title is 'Domain Controller Options'. On the left is a navigation pane with links: 'Deployment Configuration', 'Domain Controller Options' (selected), 'DNS Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main area contains the following sections:

- Select functional level of the new forest and root domain**
 - Forest functional level: Windows Server 2012 R2
 - Domain functional level: Windows Server 2012 R2
- Specify domain controller capabilities**
 - ☒ Domain Name System (DNS) server
 - ☒ Global Catalog (GC)
 - ☐ Read only domain controller (RODC)
- Type the Directory Services Restore Mode (DSRM) password**
 - Password: [masked]
 - Confirm password: [masked]

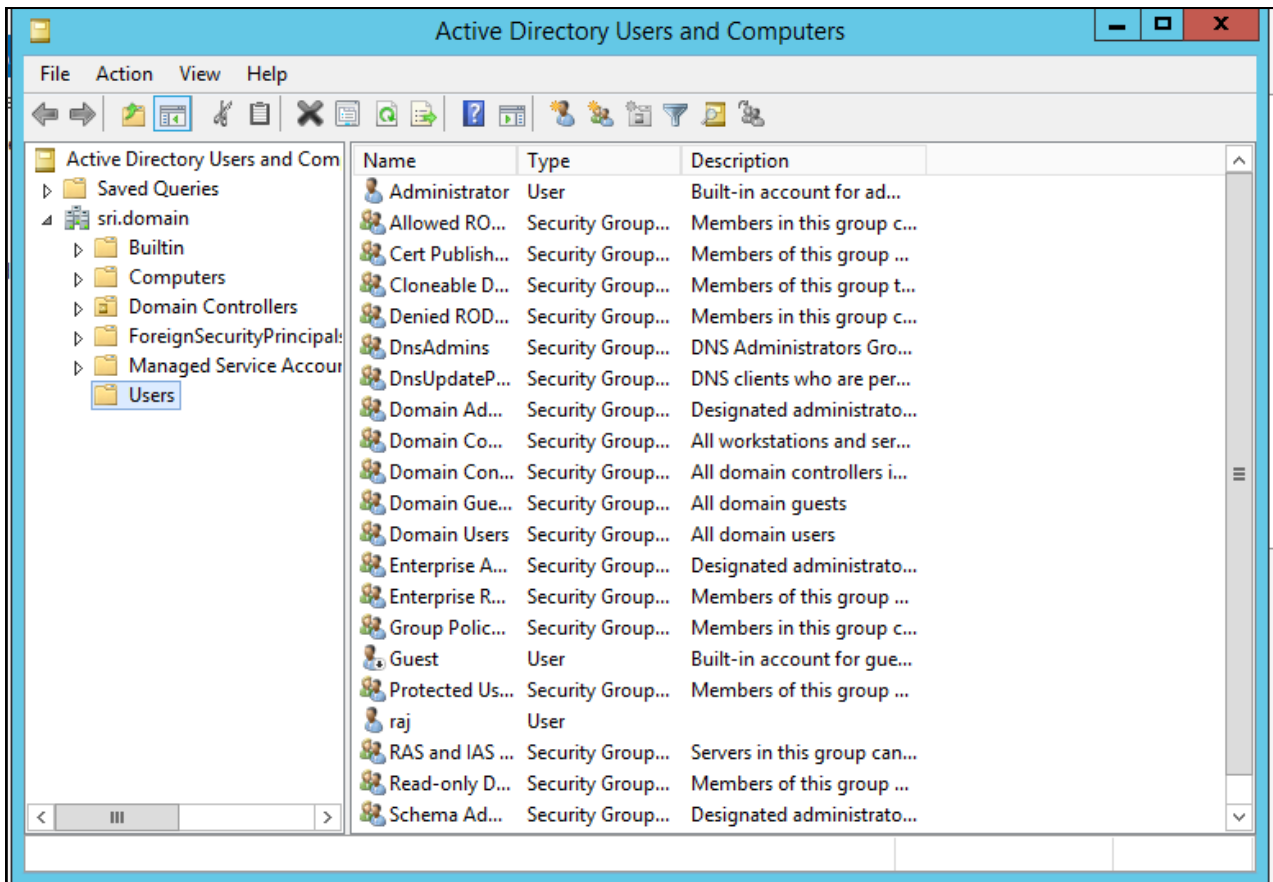
At the bottom right, it says 'TARGET SERVER WIN-2JK0M7Q654S'. At the bottom are buttons: '< Previous', 'Next >', 'Install', and 'Cancel'. A link 'More about domain controller options' is at the bottom left of the main area.

8. Log in as SRI\Administrator; AD and integrated DNS are now running.



5.Create AD users on Server :

- 1.Log in to the domain controller with a domain admin account.
- 2.Open Server Manager → Tools → Active Directory Users and Computers.
- 3.In the left pane, expand your domain

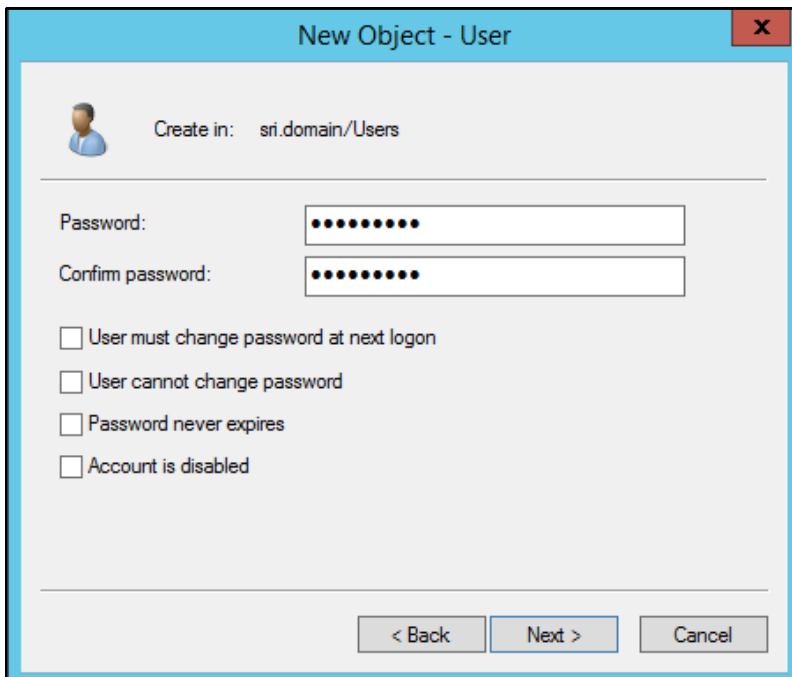


- 4.Right-click the OU/container → New → User.
- 5.Fill: First name, Last name, and User logon name (for example raj or raj@lab.local)
Next.

The 'New Object - User' dialog box is shown. The 'Create in:' field is set to 'sri.domain/Users'. The 'First name' field contains 'raj', and the 'Full name' field also contains 'raj'. The 'User logon name' field contains 'sri101' and the domain dropdown is set to '@sri.domain'. The 'User logon name (pre-Windows 2000)' field contains 'SRI\'. The 'Initials' field is empty. At the bottom, there are buttons for '< Back', 'Next >', and 'Cancel'.

6. Set a password and options:

- Usually User must change password at next logon is ticked for normal users.
- Untick User cannot change password and Password never expires for normal users.



New Object - User

Create in: sri.domain/Users

Password: [masked]

Confirm password: [masked]

☐ User must change password at next logon

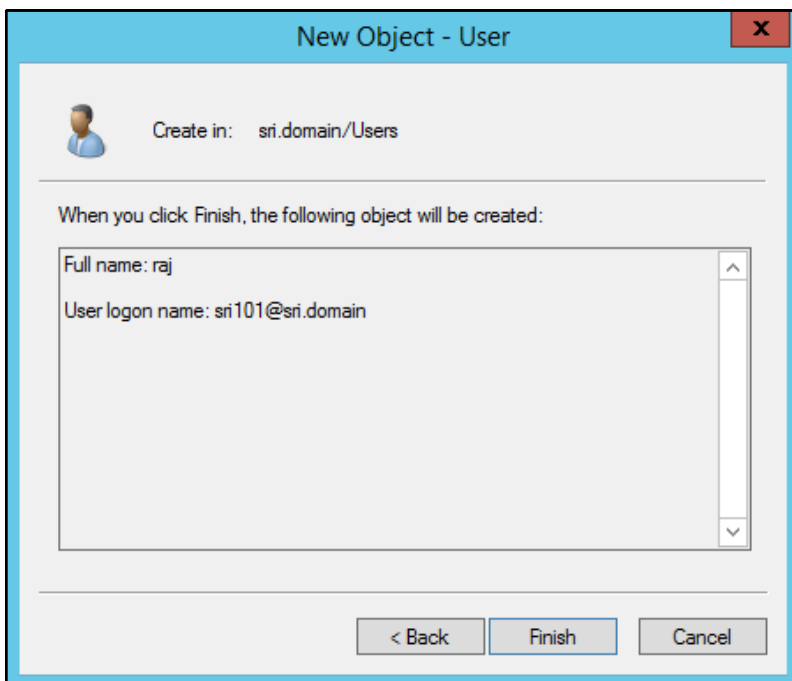
☐ User cannot change password

☐ Password never expires

☐ Account is disabled

< Back Next > Cancel

7. Finish; the new user now exists in AD and can log in on any domain-joined machine (if not restricted by policy).



New Object - User

Create in: sri.domain/Users

When you click Finish, the following object will be created:

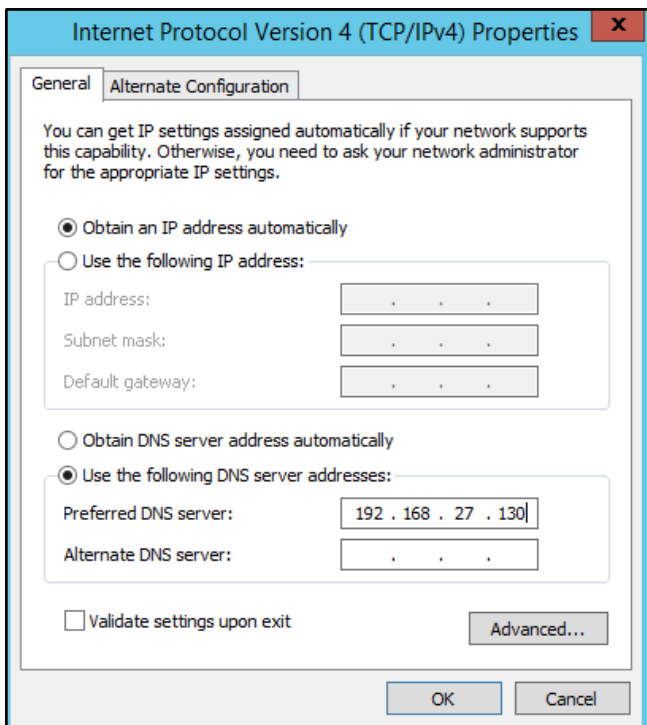
Full name: raj

User logon name: sri101@sri.domain

< Back Finish Cancel

6. Join another Windows OS to the Domain :

1. On the client, set DNS to the IP of your domain controller.

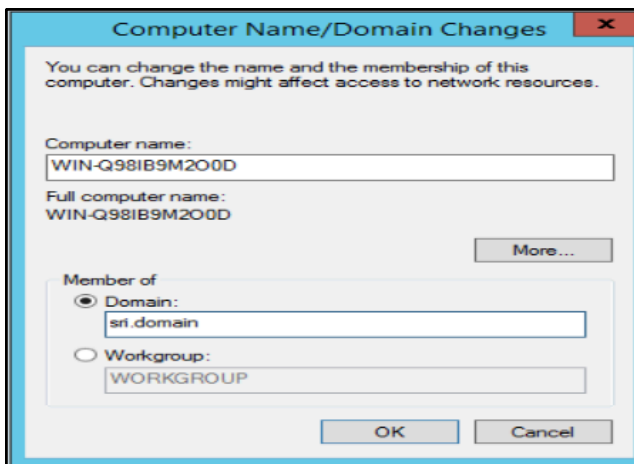


2. On the client, open:

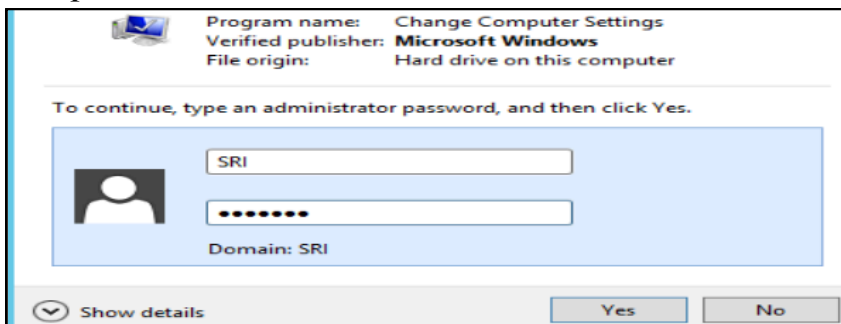
- Control Panel → System and Security → System → Change settings (right-click This PC → Properties also works).

3. Click Change... next to "To rename this computer or change its domain".

4. Under Member of, select Domain, then type your domain name.



5. When prompted, enter a domain admin account (for example SRI\Administrator) and password.



7. Log in on the client using the Domain User :

1. On the joined client's logon screen, click Other user (or choose your domain from the sign-in options).
2. Type the AD user credentials you created, for example:
 - Username: SRI\raj or raj@sri.domain.
 - Password: the password set in AD.

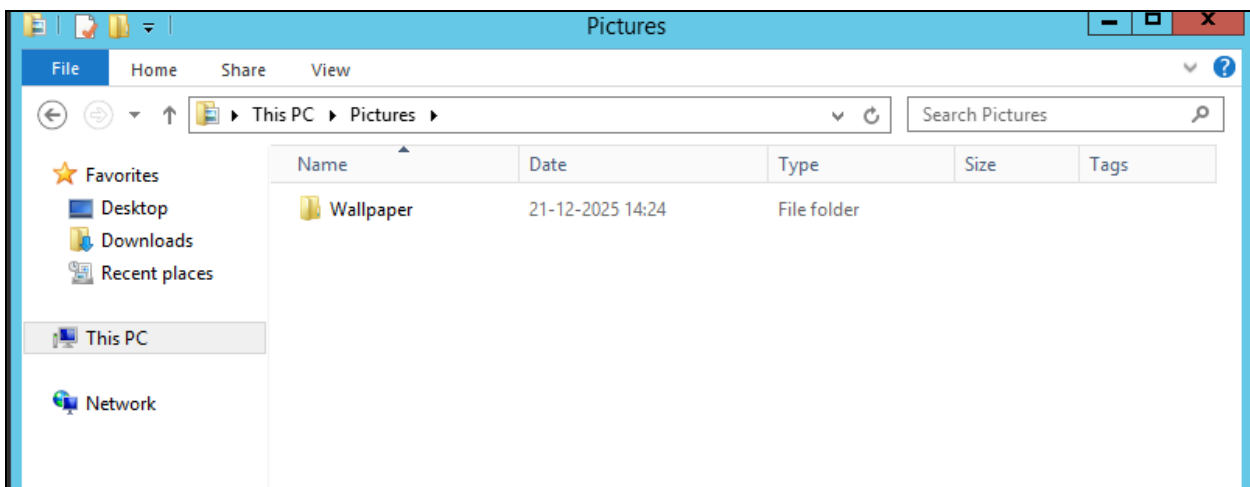


3. After first logon, Windows creates a new local profile for that domain user on that machine.

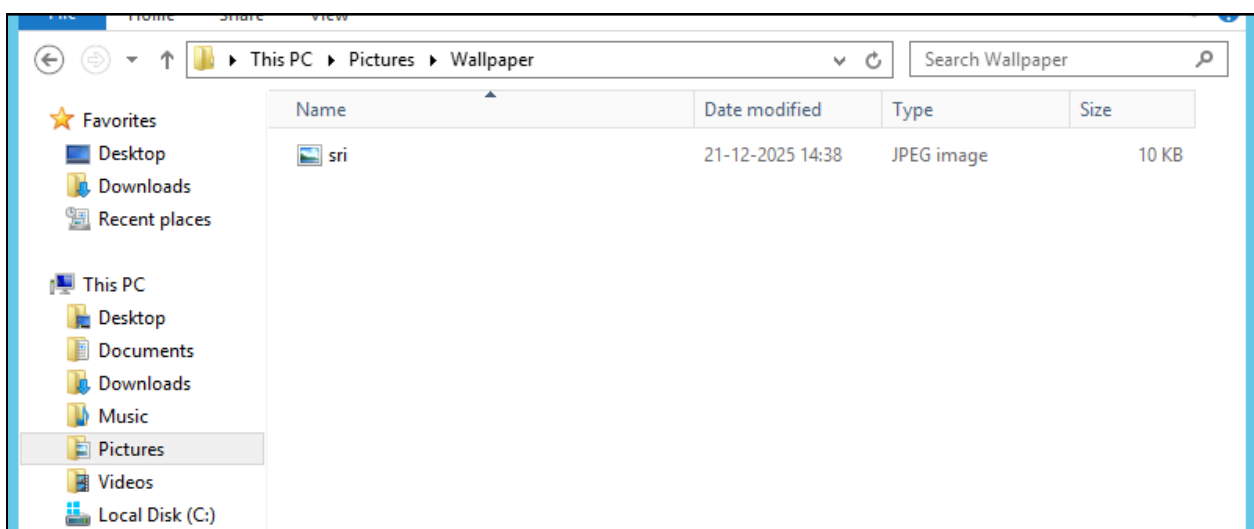
8. To Set Wallpaper On Client OS :

i. Prepare the wallpaper on the server :

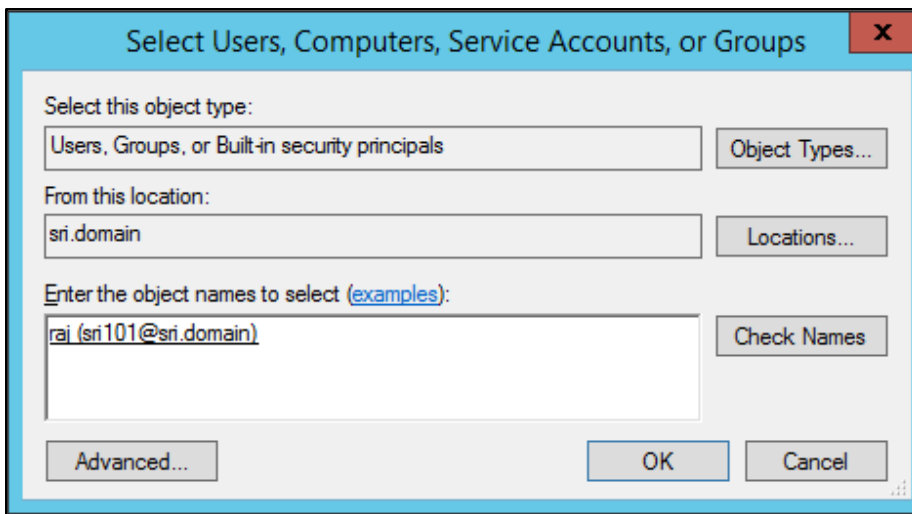
1. On the domain controller, create a folder.



2. Copy your image file (JPG/BMP) into this folder.

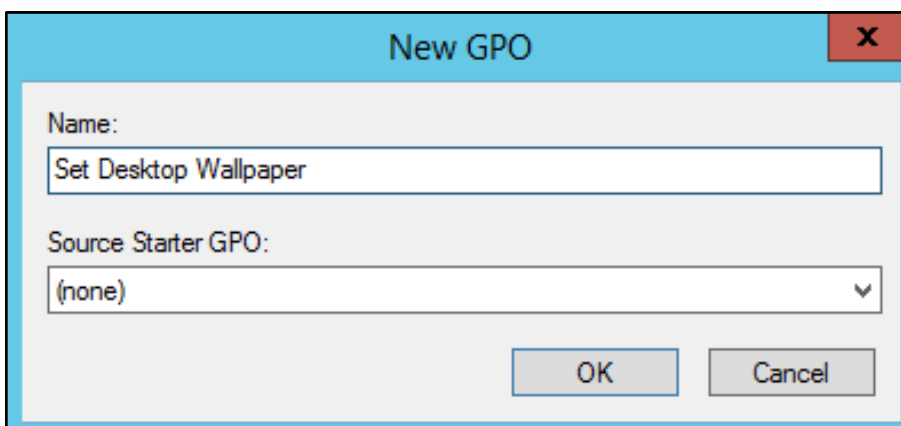


3.Right-click the folder → Properties → Sharing → Advanced Sharing → share as Wallpapers, and give Read permission to Domain Users (or the users you want).



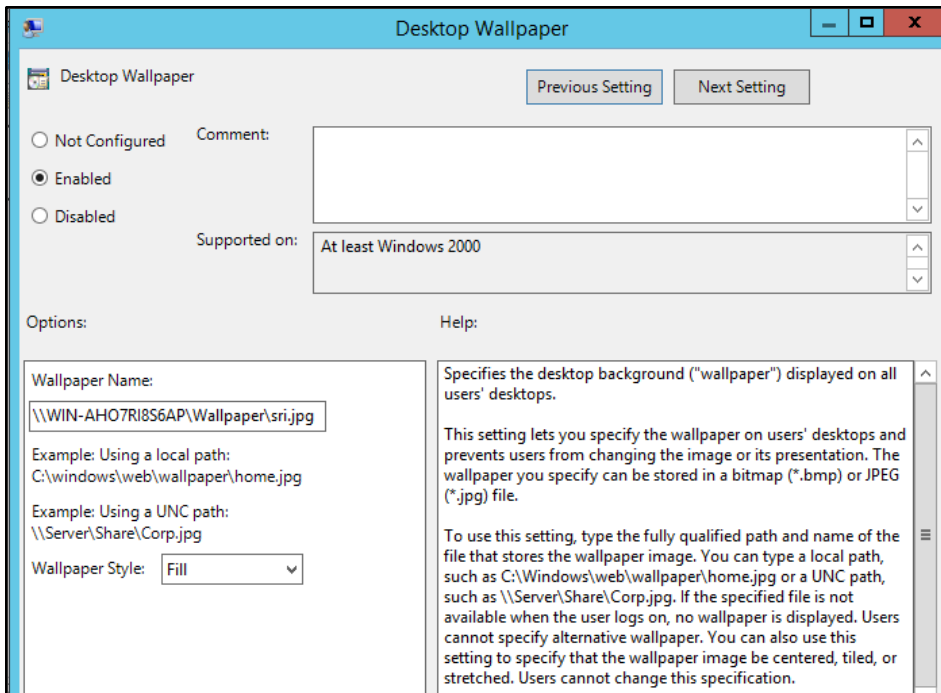
ii.Create and link the GPO :

- 1.On the DC, open Server Manager → Tools → Group Policy Management.
- 2.Right-click the OU that contains your user accounts (for example SRIUsers) → Create a GPO in this domain, and Link it here....
- 3.Name it something like Set Desktop Wallpaper → OK.



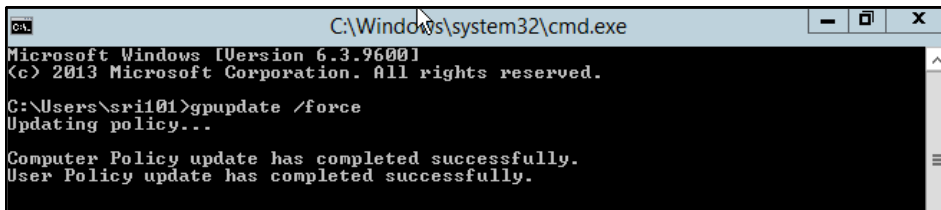
iii. Configure the wallpaper setting in the GPO :

- 1.Right-click the new GPO (Set Desktop Wallpaper) → Edit.
- 2.In the editor, go to:
 - User Configuration → Policies → Administrative Templates → Desktop → Desktop.
- 3.Double-click Desktop Wallpaper.
- 4.Set it to Enabled.
- 5.n Wallpaper name, type the UNC path to the image, for example:
 - \\DC01\\Wallpapers\\sri.jpg.
- 6.Set Wallpaper Style to Fill, Fit, or Center (usually Fill works best).
- 7.Click Apply → OK and close the editor.



iv. Apply and Test on Client OS :

1. On a domain client (Windows 10/11 joined to the domain), log in as a user from the OU where the GPO is linked.
2. Run gpupdate /force in Command Prompt, or simply log off and log on again.



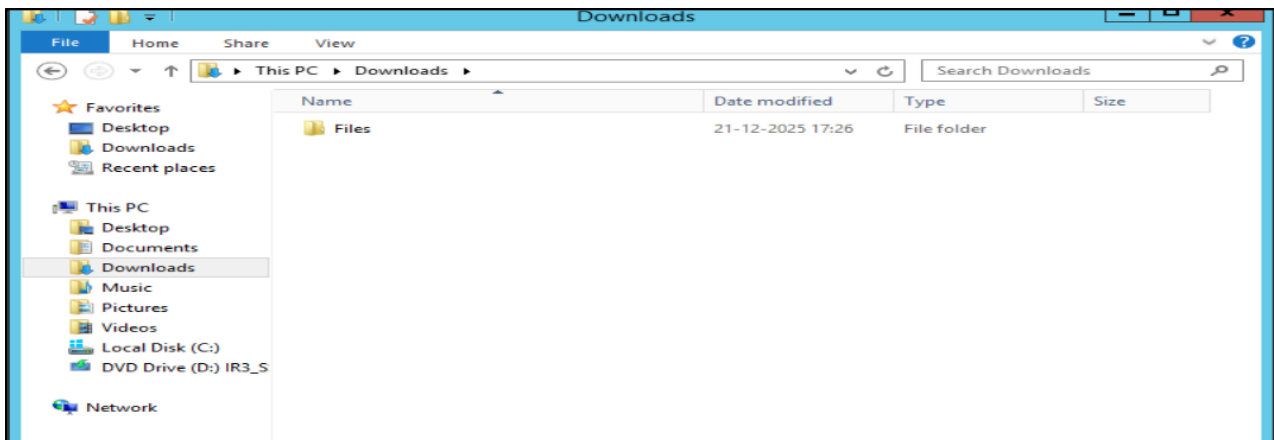
3. After policy refresh, the user's desktop background should change to the specified wallpaper.



9.File Sharing on Certain Folder :

i. Create a shared folder on the server :

1.On the server OS (for example Windows Server 2012 R2/2019), create a folder like C:\Downloads\Files.



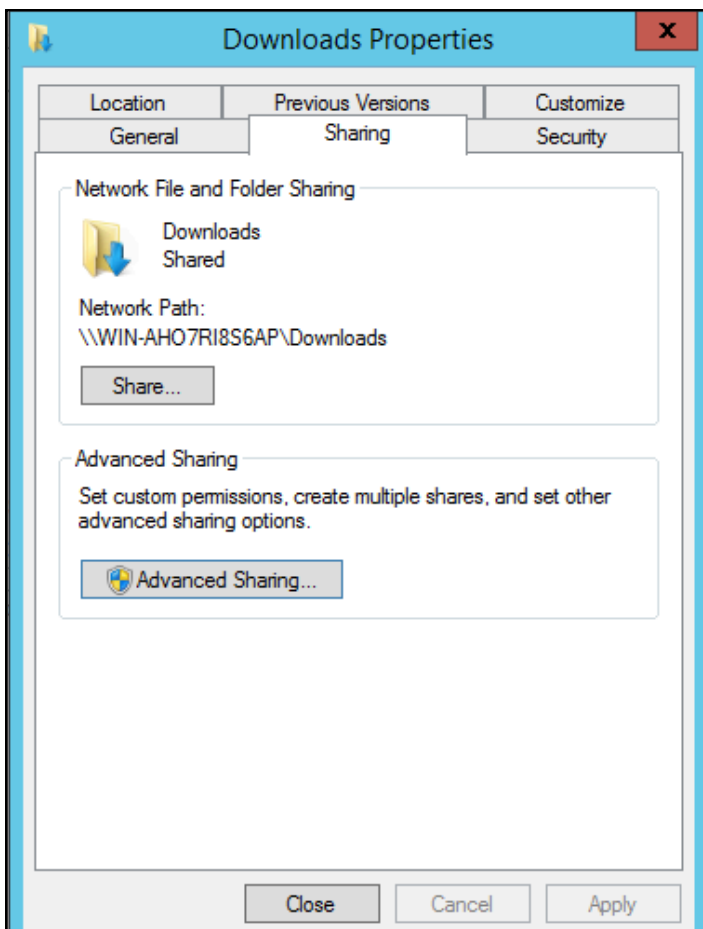
2.Right-click the folder → Properties → Sharing tab → Advanced Sharing.

3.Tick Share this folder and give it a share name, for example Files.

4.Click Permissions:

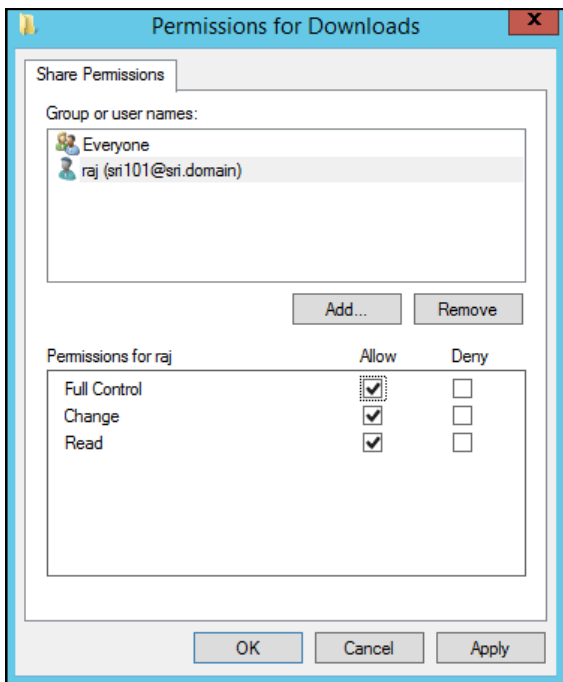
- For simple labs, allow Everyone → Full Control, then control access with NTFS permissions (next step).
- In production, restrict at share level too, but NTFS is primary.

5.Click OK → OK. The UNC path will be like \\SRV-DC01\Files.



ii. Set NTFS permissions (who can access) :

1. On the same folder, go to Properties → Security tab.
2. Click Edit (or Advanced) to adjust permissions.
3. Recommended approach:
 - Create AD security groups (for example GRP_Data_Read, GRP_Data_Modify) and add users to these groups.
 - On the folder, remove unnecessary default entries (leave SYSTEM and Administrators).
4. Add GRP_Data_Modify with Modify or Full as needed; add GRP_Data_Read with Read & execute, List folder contents, Read.
5. Apply and close; now only those group members can read/write the share.



iii. Access the share from the client OS :

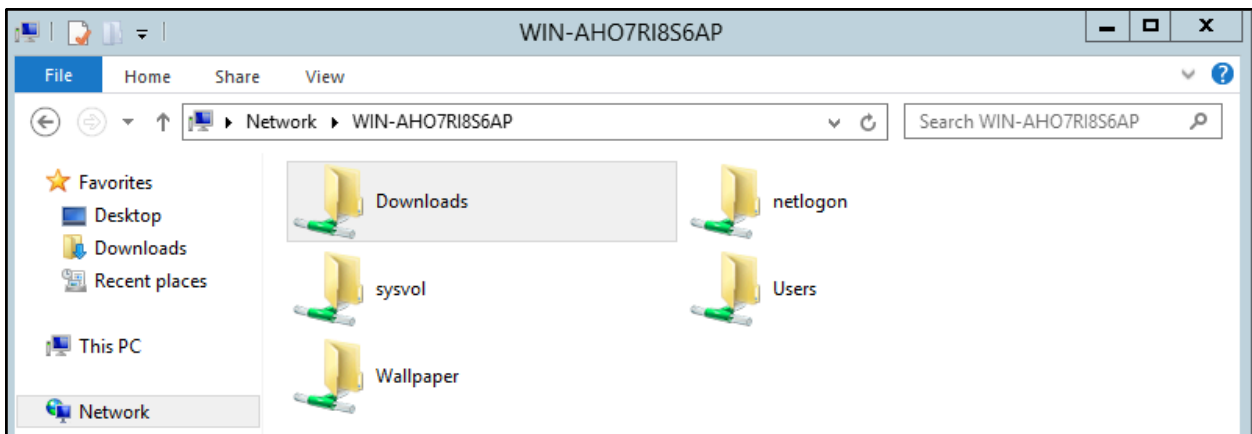
1. From a domain-joined client (Windows 10/11):
2. Ensure the client can resolve and reach the server (DNS to DC, ping works).
3. Open File Explorer → in the address bar type \\SRV-DC01\Data and press Enter.
4. When prompted, log in with a domain account that has NTFS permission; then you will see and use the shared folder.

```
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\sri101>ping 192.168.27.130

Pinging 192.168.27.130 with 32 bytes of data:
Reply from 192.168.27.130: bytes=32 time<1ms TTL=128
Reply from 192.168.27.130: bytes=32 time=1ms TTL=128
Reply from 192.168.27.130: bytes=32 time<1ms TTL=128
Reply from 192.168.27.130: bytes=32 time<1ms TTL=128

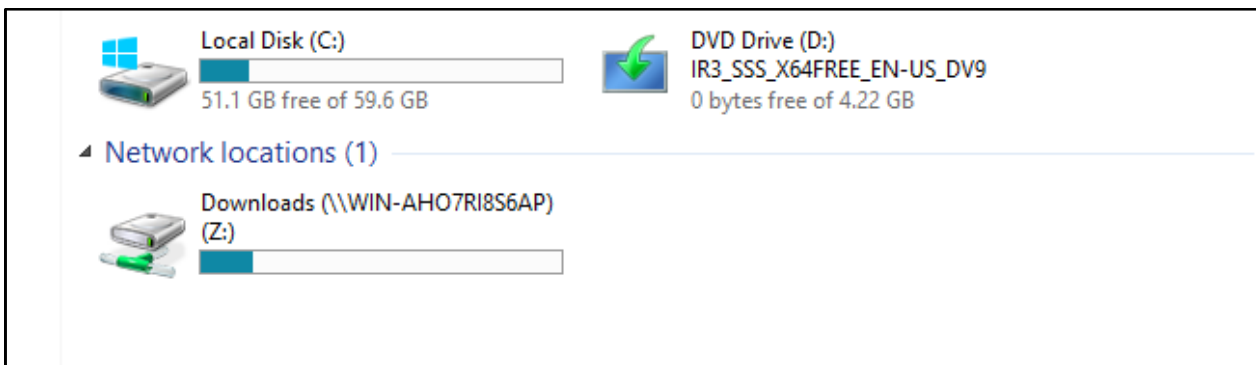
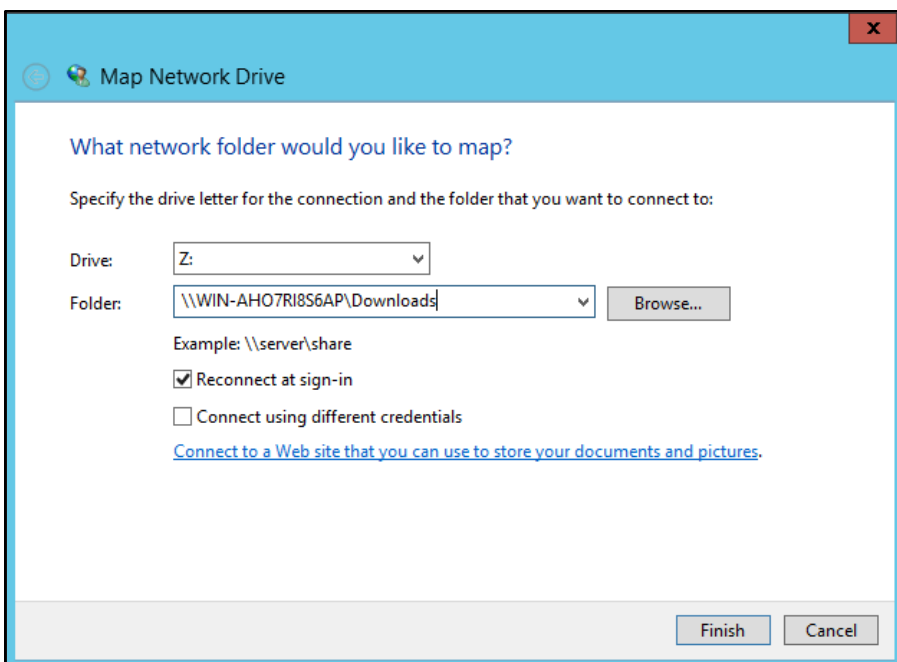
Ping statistics for 192.168.27.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```



iv. Map as a network drive (optional, user friendly):

On the client:

1. Open File Explorer → This PC → Map network drive.
2. Choose a drive letter (for example Z:).
3. In Folder, type \\SRV-DC01\Files.
4. Tick Reconnect at sign-in; click Finish.
5. Enter domain credentials if asked; the drive appears as Z: and points to the server share.



10. Configure File Server Resource Manager (FSRM) :

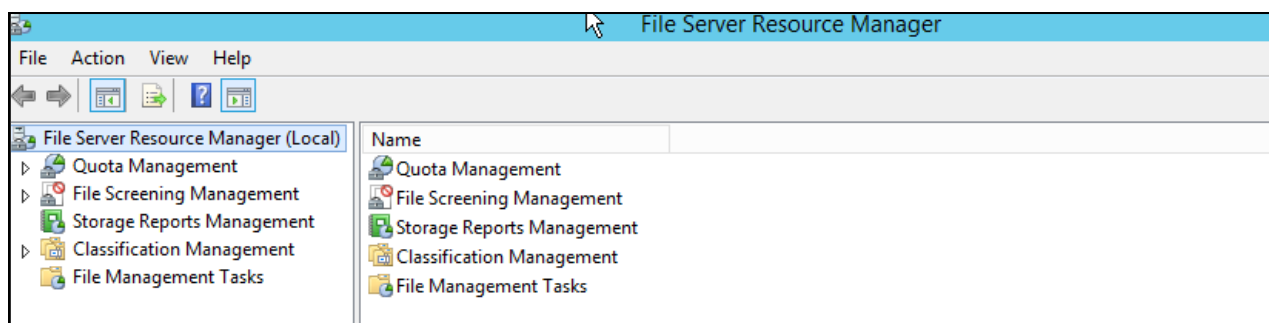
i. Install FSRM role :

1. On Windows Server 2012 R2/2019, open Server Manager → Add roles and features.
2. Role-based → select your server → File and Storage Services → File and iSCSI Services.
3. Tick File Server Resource Manager → add features → Install.
4. After install: Server Manager → Tools → File Server Resource Manager.

ii. Configure storage quotas :

Quotas limit folder space usage per user or group.

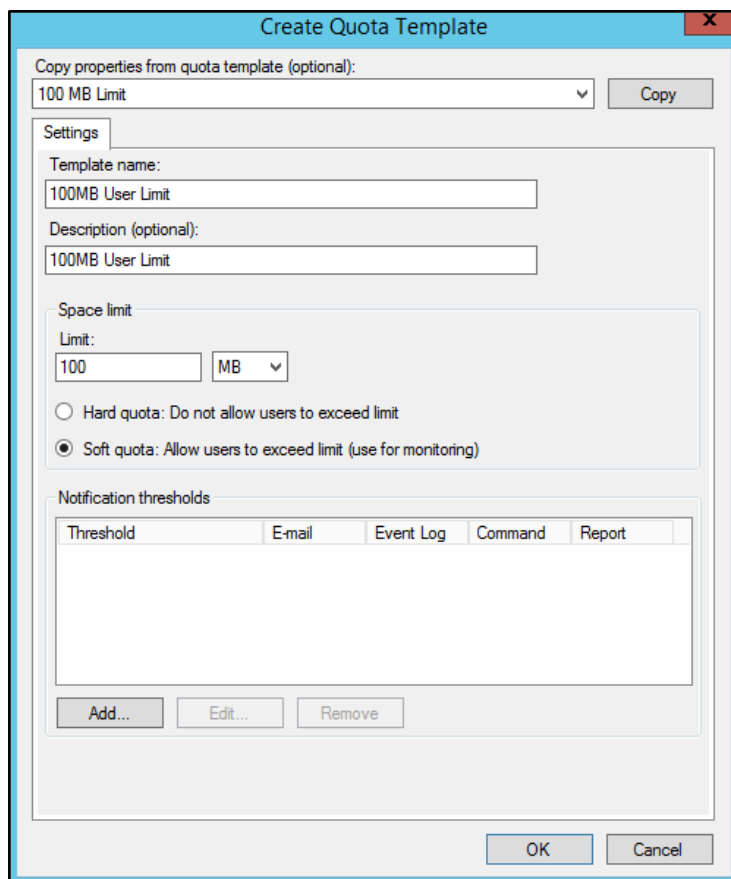
1. In FSRM, expand Quotas → right-click Quota Templates → Create Quota Template.
2. Name it (for example 50GB User Limit); set Space limit to 100 MB.



3. Notification:

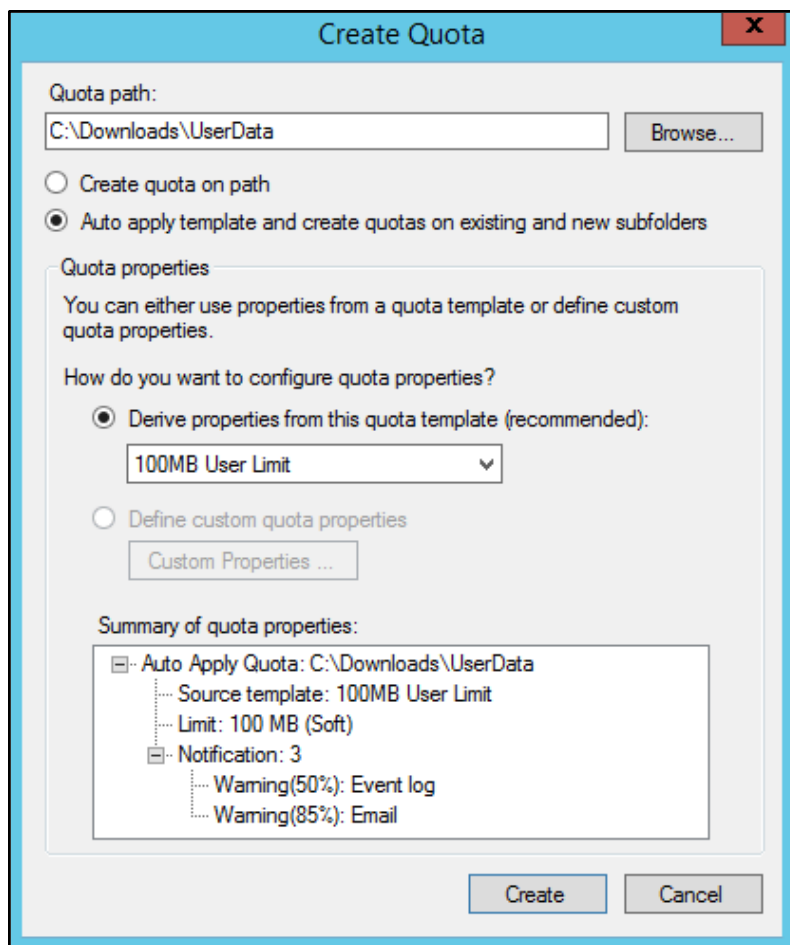
- 50% threshold → E-mail Message to admin (optional).
- 85% → Event Log + Run Script (optional).

4. Click OK to save template.



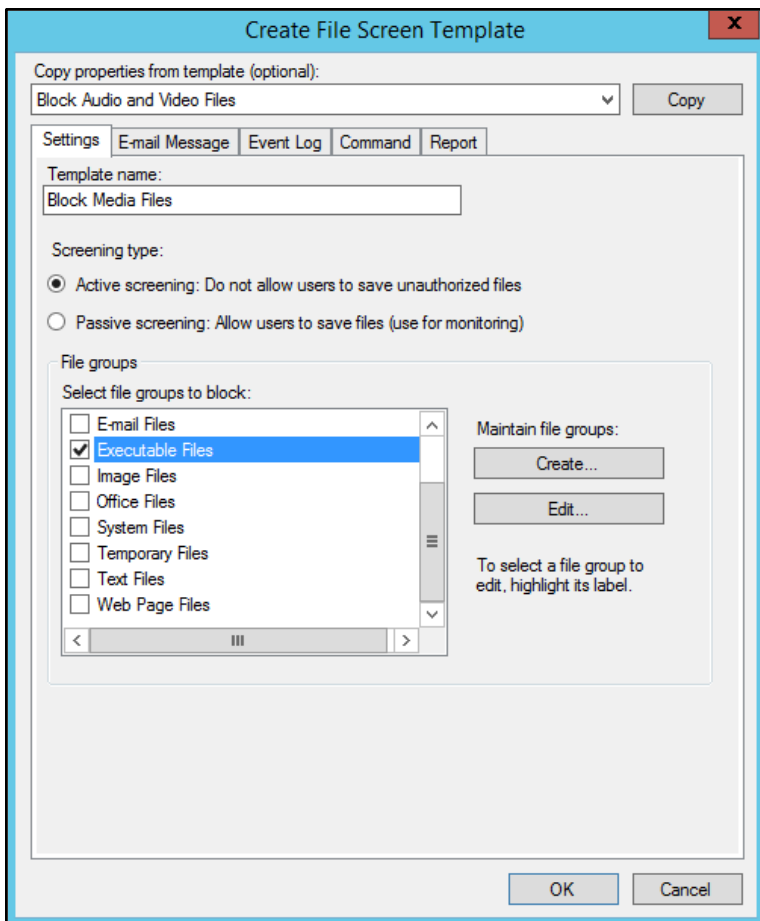
Apply quota to a folder:

- 1.Right-click Quotas → Create Quota....
- 2.Browse to folder (for example D:\Downloads\Files).
- 3.Select Auto apply template → choose 100MB User Limit → Create quota.



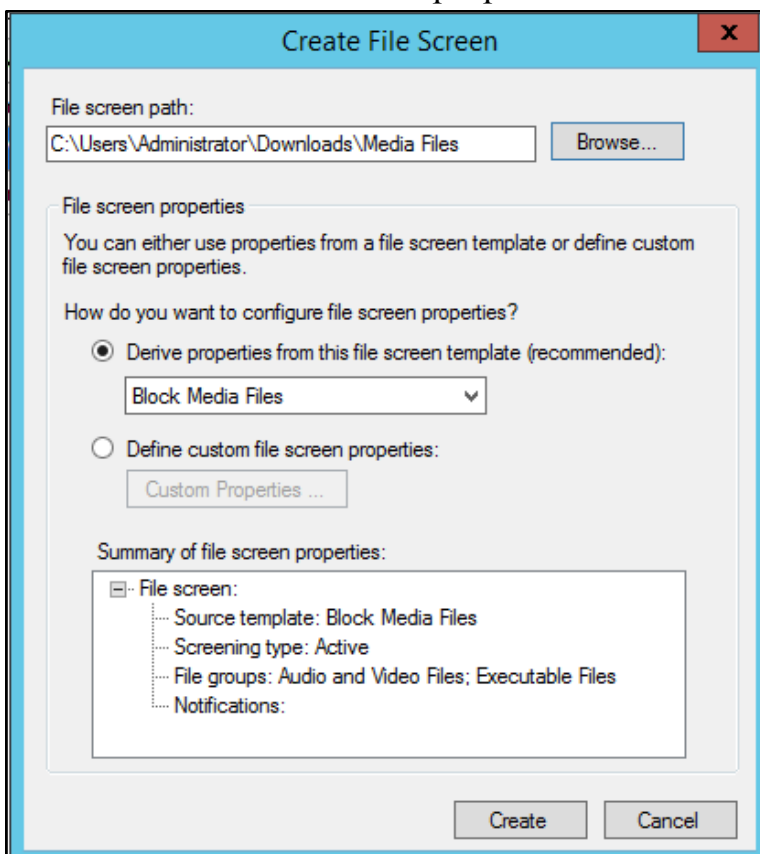
iii. Configure file screening :

- 1.Expand File Screening → right-click File Screen Templates → Create File Screen Template.
- 2.Name: Block Media Files; Screening type: Active screening (blocks).
- 3.File groups:
 - Select Audio and Video Files (blocks MP3, AVI, etc.).
 - Or Executable Files (EXE, BAT, MSI).
- 4.Notifications (optional): Event Log, E-mail.
- 5.OK to save.



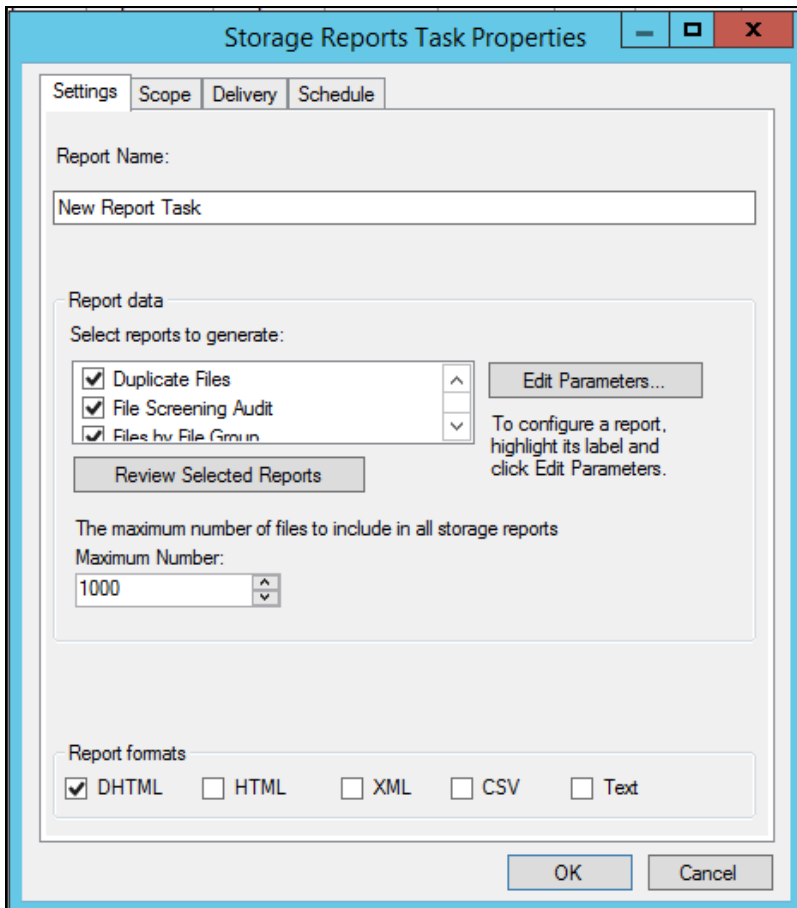
Apply screen:

1. Right-click File Screens → Create File Screen....
2. Browse to folder → Derive properties from Block Media Files template → Create.



iv. Generate storage reports :

1. In FSRM → Storage Reports Management → Schedule a New Report Task.
2. Choose report: Large Files, Most Files, Quota Usage, File Screen Activity.
3. Select Parameters (for example top 10 largest files >1GB).
4. Delivery: Save to C:\Reports or e-mail.
5. Schedule daily/weekly → OK.



v. Verify Quotas and Screens :

1. Quota status: FSRM → Quotas → right-click quota → Properties (shows usage).
2. Screen events: Event Viewer → Applications and Services → Microsoft → Windows → File Server Resource Manager.
3. Test: Try copying a large file or MP3 to a screened folder → should be blocked.

Filter: Show all: 1 items						
Quota Path	% Us...	Limit	Quota T...	Source Template	Match Tem...	Description
Source Template: 100MB User Limit (1 item)						
C:\Users\Administr...	---	100 ...	Soft (Aut...	100MB User Limit	Yes	

Filter: Show all: 1 items					
File Screen Path	Screening Ty...	File Groups	Source Template	Match...	
Source Template: Block Media Files (1 item)					
C:\Users\Administrat...	Active	Block: Audio and Video ...	Block Media Files	Yes	

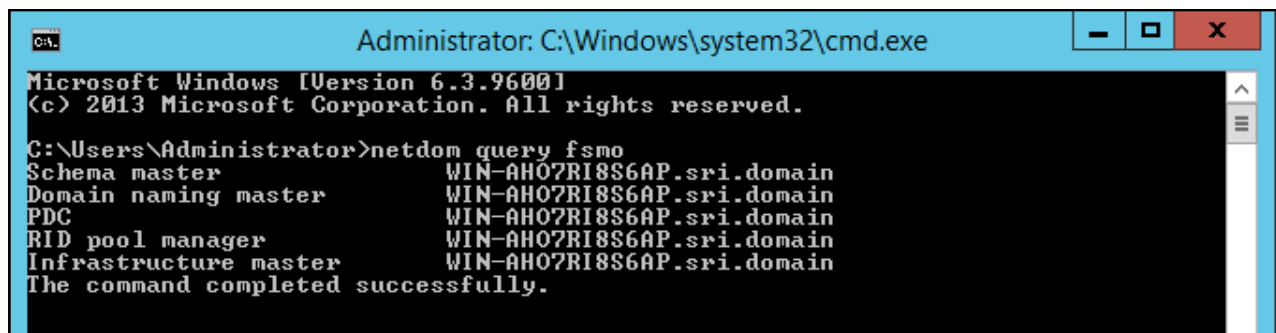
11.To Configure FSMO Rules :

i. Know the 5 FSMO roles :

- 1.Forest-wide (1 per forest): Schema Master, Domain Naming Master.
- 2.Domain-wide (1 per domain): RID Master, PDC Emulator, Infrastructure Master.
3. Check current FSMO role holders
 - On any DC (Server 2012 R2/2016/2019):
- 4.Open Command Prompt as Administrator → run:

Text :

netdom query fsmo



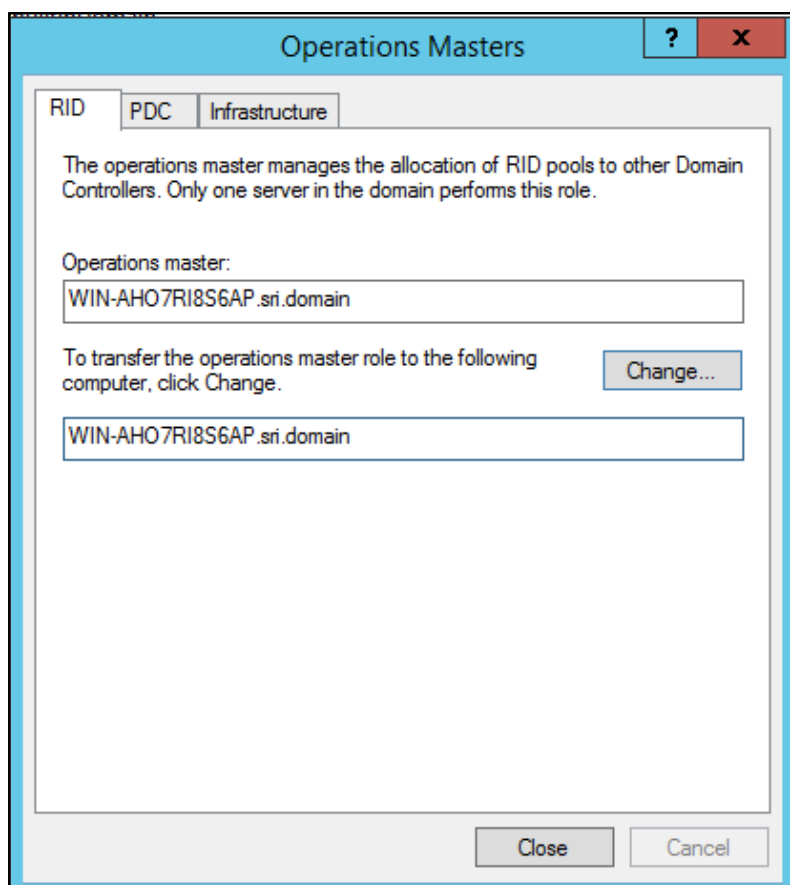
```
Administrator: C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>netdom query fsmo
Schema master           WIN-AHO7RI8S6AP.sri.domain
Domain naming master    WIN-AHO7RI8S6AP.sri.domain
PDC                     WIN-AHO7RI8S6AP.sri.domain
RID pool manager        WIN-AHO7RI8S6AP.sri.domain
Infrastructure master    WIN-AHO7RI8S6AP.sri.domain
The command completed successfully.
```

ii.Transfer roles with GUI (normal, both DCs online) :

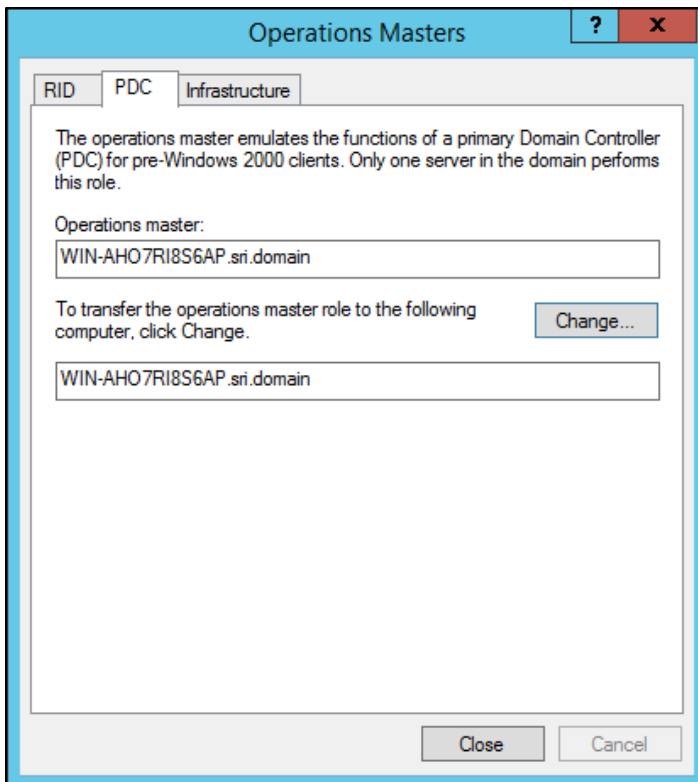
a) RID, PDC, Infrastructure (AD Users and Computers)

- 1.On DC2, open Active Directory Users and Computers.
- 2.Right-click the domain name → Operations Masters....
- 3.Use each tab: RID, PDC, Infrastructure → for each, click Change → Yes.



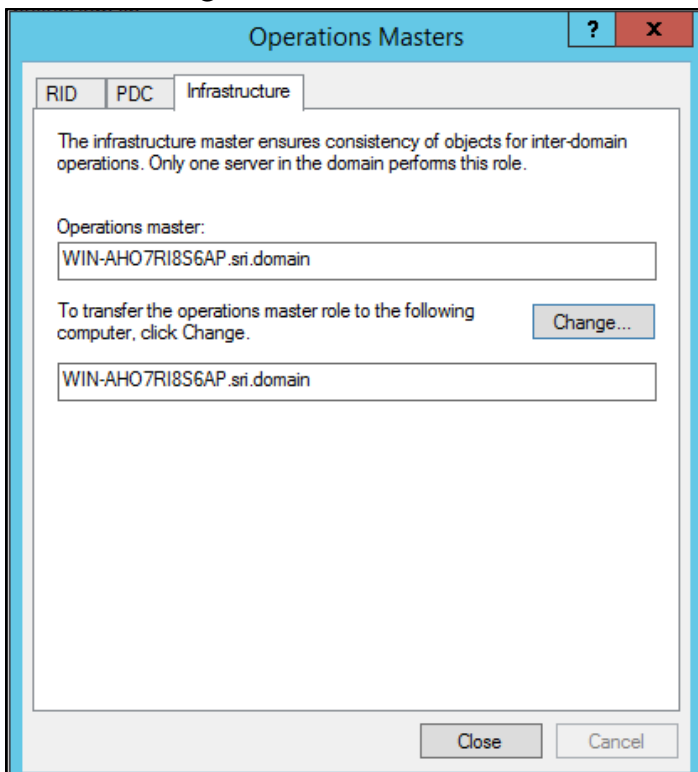
b) Domain Naming Master (AD Domains and Trusts)

1. On DC2, open Active Directory Domains and Trusts.
2. Right-click Active Directory Domains and Trusts (top) → Operations Master....
3. Click Change → Yes.



c) Infrastructure (AD Domains and Trusts)

1. On DC2, open Active Directory Domains and Trusts.
2. Right-click Active Directory Domains and Trusts (top) → Operations Master....
3. Click Change → Yes.



d) Schema Master (AD Schema snap-in)

1. On DC2, run:

Text:

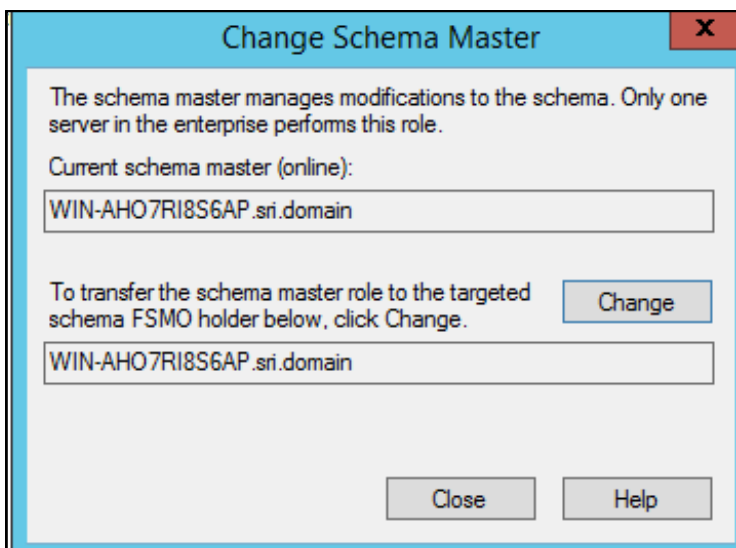
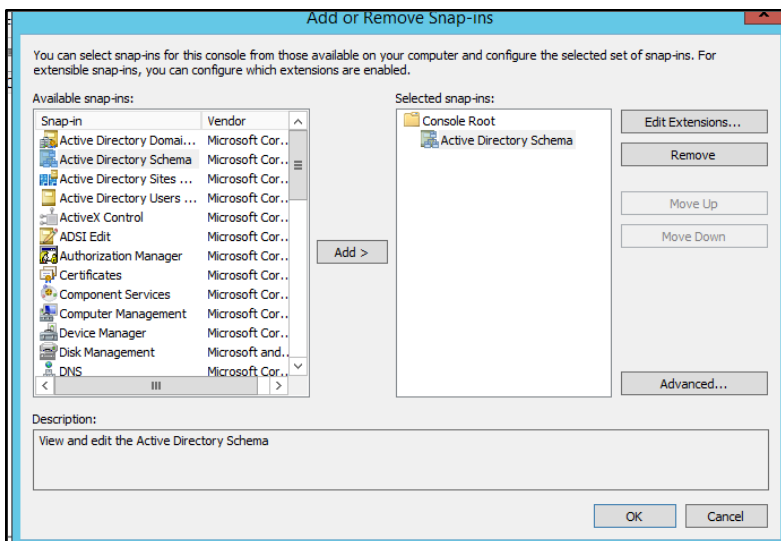
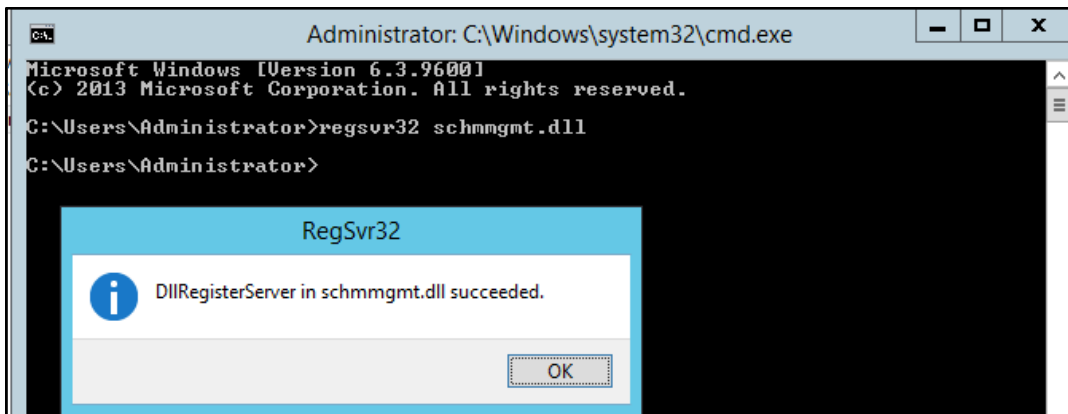
- `regsvr32 schmmgmt.dll`

2. then open MMC → Add/Remove Snap-in → Active Directory Schema.

3. Right-click Active Directory Schema → Operations Master....

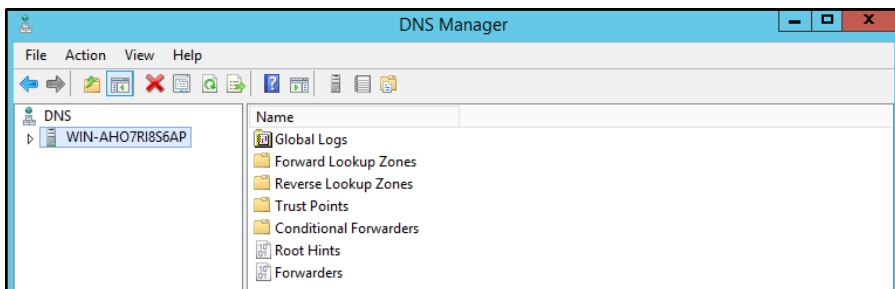
4. Click Change → Yes.

- Re-run `netdom query fsmo` to confirm roles now show DC2.



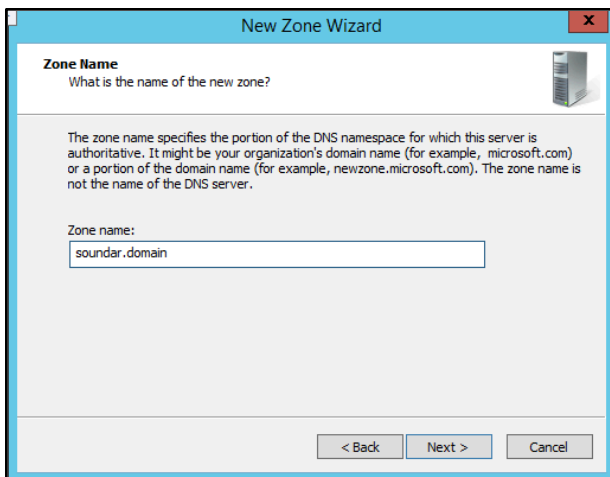
12.Configure DNS on the Windows Server OS :

- 1.On the server, open Server Manager → Add roles and features.
- 2.Role-based installation → select this server → tick DNS Server → Next → Install.
- 3.After install, open Tools → DNS to launch DNS Manager.



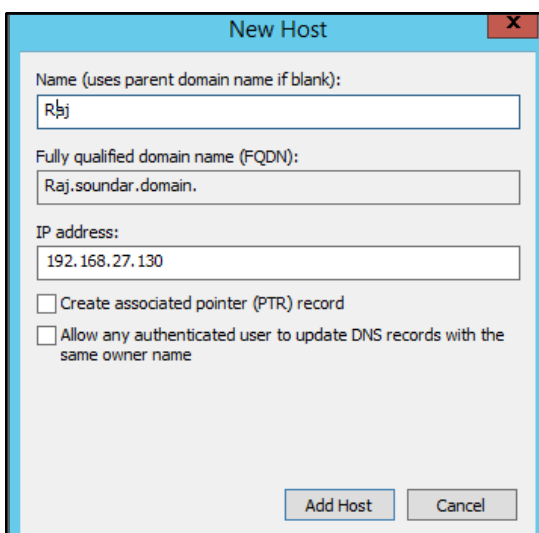
a).Create or Check your Zone :

- 1.In DNS Manager, expand your server → Forward Lookup Zones.
- 2.If your domain zone (for example lab.local) exists from AD, use it; otherwise right-click Forward Lookup Zones → New Zone..., create a Primary zone, name lab.local, store in AD if available.



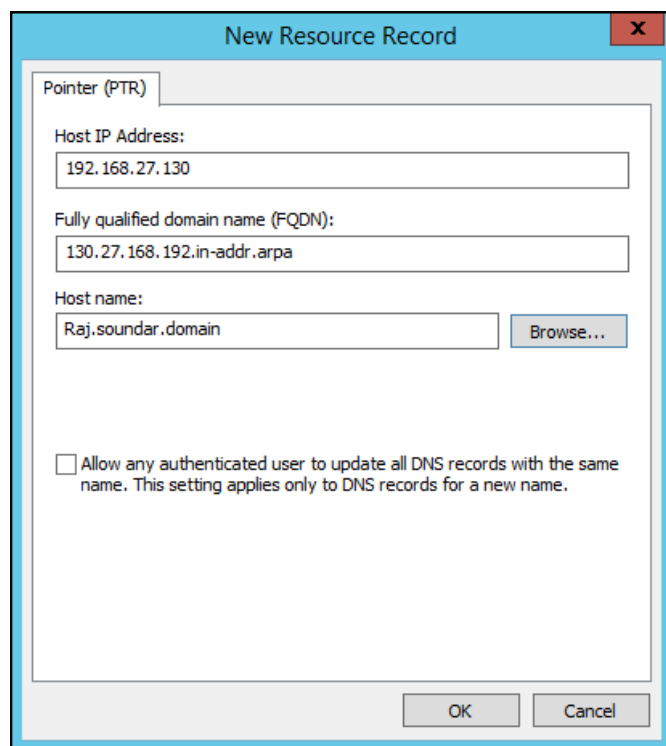
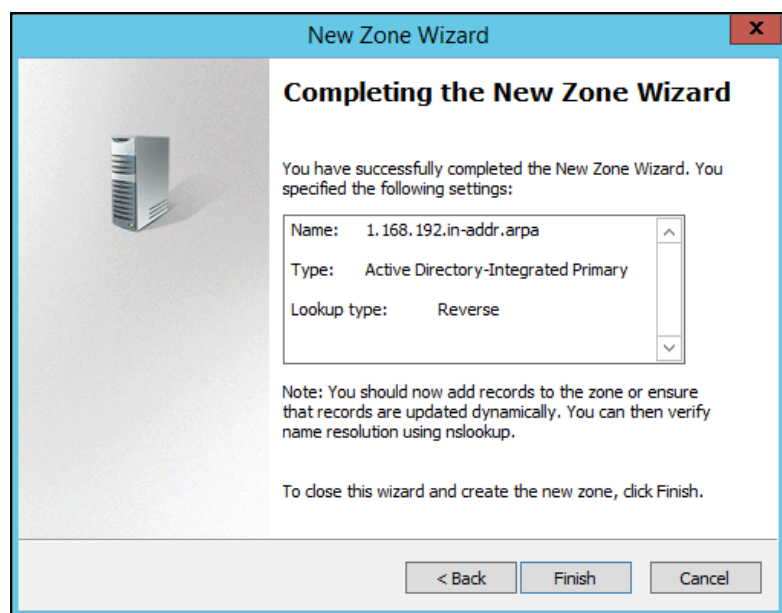
- 3.Right-click the zone → New Host (A or AAAA)... and create records for important hosts:

- Name: blank or @, IP: server IP (for example 192.168.27.130) → creates soundar.domain.
- Name: dc1, IP: 192.168.27.130 → creates Raj.soundar.domain.



b).Reverse lookup zone:

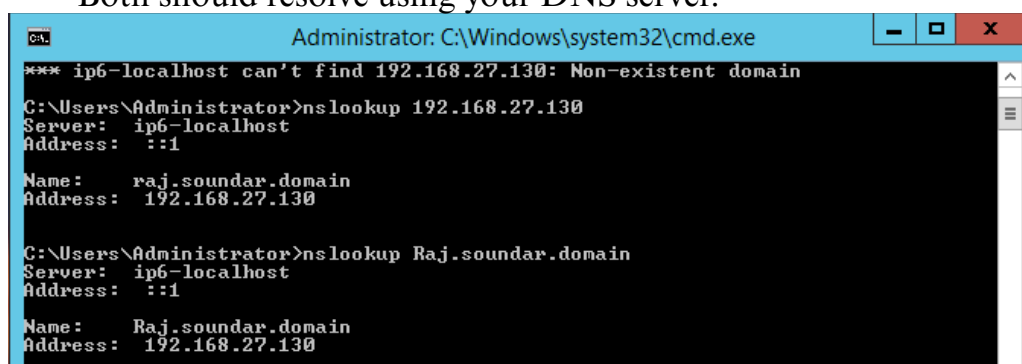
- 1.Right-click Reverse Lookup Zones → New Zone..., choose network ID (for example 192.168.27).
- 2.Tick “Create associated PTR record” when adding A records so reverse lookups work.



c).Test on the server:

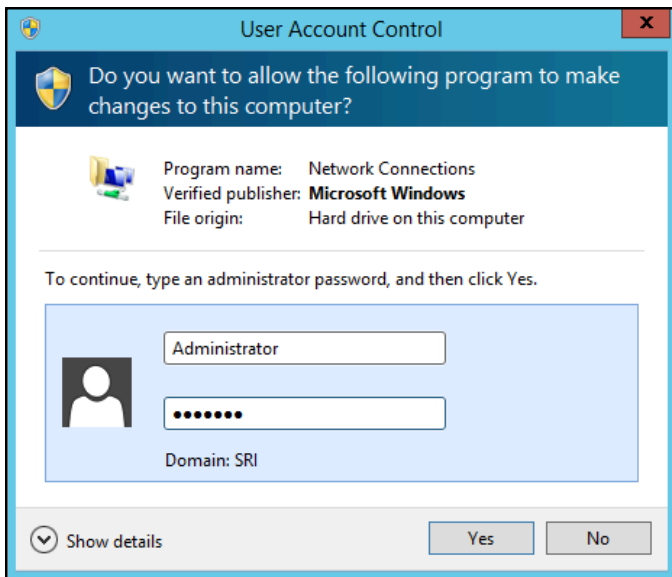
- 1.Open CMD →
 - nslookup dc1.lab.local
 - nslookup 192.168.1.10

Both should resolve using your DNS server.



13. Configure DNS on the Client OS (Manual):

1. On the client, open Control Panel → Network and Internet → Network and Sharing Center → Change adapter settings.
2. Right-click your active adapter (Ethernet/Wi-Fi) → Properties.



3. Select Internet Protocol Version 4 (TCP/IPv4) → Properties.
4. Choose Use the following DNS server addresses.
5. In Preferred DNS server, enter your server's IP (for example 192.168.27.130). Leave Alternate DNS empty or set to another internal DNS if you have it.
6. Click OK → Close on all windows.

Verify from Client:

- Open CMD → ipconfig /all and confirm DNS server = 192.168.27.130.
- Run nslookup Raj.soundar.domain; it should answer from your server.

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.3.9600]
(c) 2013 Microsoft Corporation. All rights reserved.

C:\Users\sri101>ipconfig /all

Windows IP Configuration

Host Name . . . . . : WIN-Q98IB9M200D
Primary Dns Suffix . . . . . : sri.domain
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : sri.domain
                                localdomain

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-C1-34-EE
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::c8d3:a2b7:567a:502dz12<Preferred>
    IPv4 Address. . . . . : 192.168.27.131(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : 23 December 2025 00:38:18
    Lease Expires . . . . . : 23 December 2025 01:23:18
    Default Gateway . . . . . : 192.168.27.2
    DHCP Server . . . . . : 192.168.27.254
    DHCPv6 Iaid . . . . . : 301993001
    DHCPv6 Client DUID. . . . . : 00-01-00-01-30-D8-29-21-00-0C-29-C1-34-EE

    DNS Servers . . . . . : 192.168.27.130
    Primary WINS Server . . . . . : 192.168.27.2
    NetBIOS over Tcpip. . . . . : Enabled

Tunnel adapter isatap.localdomain:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : localdomain
    Description . . . . . : Microsoft ISATAP Adapter
    Physical Address. . . . . : 00-00-00-00-00-00-E0
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes

C:\Users\sri101>
```

```
Administrator: C:\Windows\system32\cmd.exe
*** ip6-localhost can't find 192.168.27.130: Non-existent domain
C:\Users\Administrator>nslookup 192.168.27.130
Server: ip6-localhost
Address: ::1

Name: raj.soundar.domain
Address: 192.168.27.130

C:\Users\Administrator>nslookup Raj.soundar.domain
Server: ip6-localhost
Address: ::1

Name: Raj.soundar.domain
Address: 192.168.27.130
```

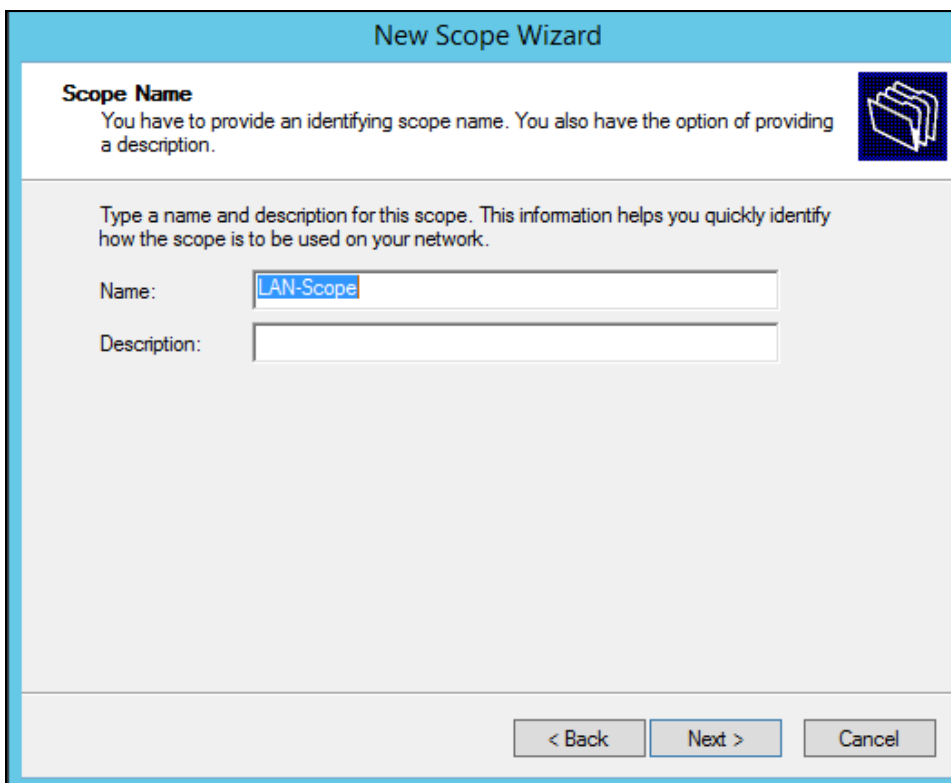
14.Install DHCP role on the Server :

- 1.Log in to your server OS with an admin account.
- 2.Open Server Manager → Add roles and features.
- 3.Role-based installation → select your server.
- 4.On Server Roles, tick DHCP Server → Add Features → Next → Next → Install.
- 5.When installation finishes, click the yellow flag in Server Manager → Complete DHCP configuration → use domain credentials to authorize the server → Commit.

i. Create a DHCP IPv4 Scope :

- 1.Open Server Manager → Tools → DHCP.
- 2.Expand your server → IPv4 → right-click IPv4 → New Scope....

Name: LAN-Scope



New Scope Wizard

Scope Name
You have to provide an identifying scope name. You also have the option of providing a description.

Type a name and description for this scope. This information helps you quickly identify how the scope is to be used on your network.

Name:

Description:

< Back Next > Cancel

3.IP range:

- Start: 192.168.27.1
- End: 192.168.27.200

New Scope Wizard

IP Address Range
You define the scope address range by identifying a set of consecutive IP addresses.

Configuration settings for DHCP Server

Enter the range of addresses that the scope distributes.

Start IP address: 192 . 168 . 27 . 1

End IP address: 192 . 168 . 27 . 200

Configuration settings that propagate to DHCP Client

Length: 24

Subnet mask: 255 . 255 . 255 . 0

4. Add Exclusions for static devices (for example 192.168.27.100–192.168.27.200 for routers/servers).

New Scope Wizard

Add Exclusions and Delay
Exclusions are addresses or a range of addresses that are not distributed by the server. A delay is the time duration by which the server will delay the transmission of a DHCP OFFER message.

Type the IP address range that you want to exclude. If you want to exclude a single address, type an address in Start IP address only.

Start IP address: 192 . 168 . 27 . 100 End IP address: 192 . 168 . 27 . 200

Excluded address range:

Subnet delay in milli second:

0

5. Lease duration: keep default (8 days) for lab.

6. When asked “Configure DHCP options now?”, choose Yes.

New Scope Wizard

Lease Duration
The lease duration specifies how long a client can use an IP address from this scope.

Lease durations should typically be equal to the average time the computer is connected to the same physical network. For mobile networks that consist mainly of portable computers or dial-up clients, shorter lease durations can be useful. Likewise, for a stable network that consists mainly of desktop computers at fixed locations, longer lease durations are more appropriate.

Set the duration for scope leases when distributed by this server.

Limited to:

Days: Hours: Minutes:

7.Scope options wizard:

- Router (Default Gateway): add 192.168.1.1

New Scope Wizard

Router (Default Gateway)
You can specify the routers, or default gateways, to be distributed by this scope.

To add an IP address for a router used by clients, enter the address below.

IP address:

8.DNS Servers: add your DNS/server IP, for example 192.168.27.139.

9.DNS Domain Name: your domain, for example sri.domain.

- Finish the wizard, then right-click the new scope → Activate.

New Scope Wizard

Domain Name and DNS Servers
 The Domain Name System (DNS) maps and translates domain names used by clients on your network.

You can specify the parent domain you want the client computers on your network to use for DNS name resolution.

Parent domain:

To configure scope clients to use DNS servers on your network, enter the IP addresses for those servers.

Server name:	IP address:	
sri.domain	192 . 168 . 27 . 139	Add
Resolve	192.168.27.130	Remove
		Up
		Down

15. Configure the client OS to use DHCP :

1. Open Control Panel → Network and Sharing Center → Change adapter settings.
2. Right-click the active adapter (Ethernet/Wi-Fi) → Properties.
3. Select Internet Protocol Version 4 (TCP/IPv4) → Properties.
4. Choose:
 - Obtain an IP address automatically.
 - Obtain DNS server address automatically.
5. Click OK → Close.

Internet Protocol Version 4 (TCP/IPv4) Properties ✕

General Alternate Configuration

You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.

☒ Obtain an IP address automatically

☐ Use the following IP address:

IP address:	. . .
Subnet mask:	. . .
Default gateway:	. . .

☒ Obtain DNS server address automatically

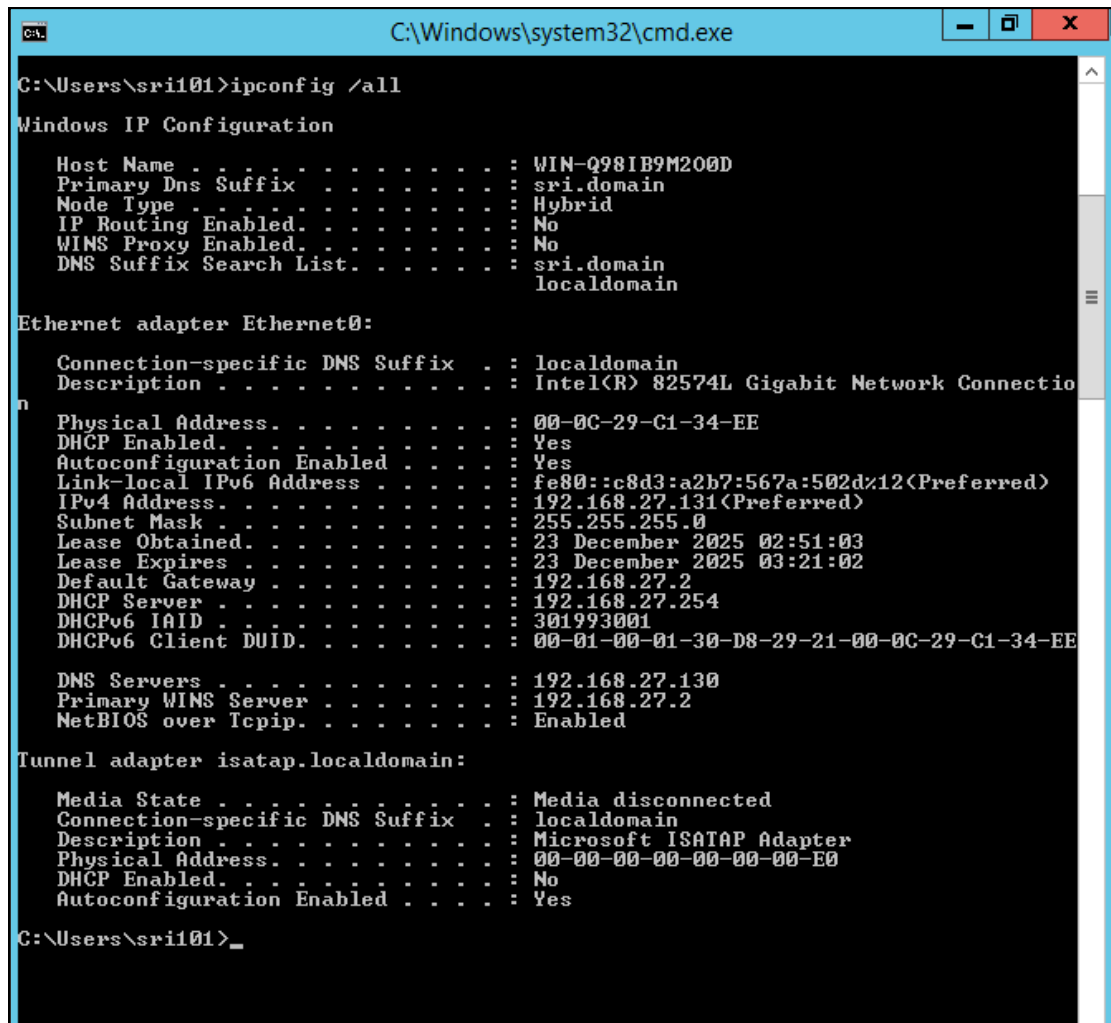
☐ Use the following DNS server addresses:

Preferred DNS server:	. . .
Alternate DNS server:	. . .

☐ Validate settings upon exit

Renew and Verify :

- 1.Open Command Prompt on the client.
- 2.Run:
 - ipconfig /release
 - ipconfig /renew
- 3.Run ipconfig /all and confirm:
- 4.IP is in your scope range (for example 192.168.27.100–200).
- 5.Default Gateway = 192.168.1.2.
- 6.DNS Server = 192.168.1.130 (your server).



```
C:\Windows\system32\cmd.exe

C:\Users\sri101>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : WIN-Q98IB9M200D
    Primary Dns Suffix . . . . . : sri.domain
    Node Type . . . . . : Hybrid
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No
    DNS Suffix Search List. . . . . : sri.domain
                                        localdomain

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix . : localdomain
    Description . . . . . : Intel(R) 82574L Gigabit Network Connection
    Physical Address. . . . . : 00-0C-29-C1-34-EE
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::c8d3:a2b7:567a:502d%12(Preferred)
    IPv4 Address. . . . . : 192.168.27.131(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Lease Obtained. . . . . : 23 December 2025 02:51:03
    Lease Expires . . . . . : 23 December 2025 03:21:02
    Default Gateway . . . . . : 192.168.27.2
    DHCP Server . . . . . : 192.168.27.254
    DHCPv6 IAID . . . . . : 301993001
    DHCPv6 Client DUID. . . . . : 00-01-00-01-30-D8-29-21-00-0C-29-C1-34-EE

    DNS Servers . . . . . : 192.168.27.130
    Primary WINS Server . . . . . : 192.168.27.2
    NetBIOS over Tcpip. . . . . : Enabled

Tunnel adapter isatap.localdomain:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : localdomain
    Description . . . . . : Microsoft ISATAP Adapter
    Physical Address. . . . . : 00-00-00-00-00-00-00-E0
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes

C:\Users\sri101>
```

TASK-2

1.Linux Administration :

Linux :

- Linux is a free, open-source operating system that sits between your hardware and applications and manages resources like CPU, memory, storage, and devices.
- It is Unix-like, very stable, secure, and powers everything from phones and laptops to most servers, supercomputers, and cloud infrastructure.

2.Linux File Types :

Type	Symbol	Description	Examples
Regular	-	Stores data like text, binaries, images, or scripts.	script.sh, image.jpg, executables
Directory	d	Contains other files and subdirectories.	/home, /etc
Symbolic Link	l	Shortcut pointing to another file or directory.	ln -s target linkname
Block Special	b	Represents block devices for buffered I/O (e.g., disks).	/dev/sda
Character Special	c	Represents character devices for stream I/O (e.g., terminals).	/dev/tty, /dev/null
Named Pipe (FIFO)	p	Enables inter-process communication via named pipes.	mkfifo pipefile
Socket	s	Facilitates network or local inter-process communication.	Unix domain sockets

3.Linux File Formats :

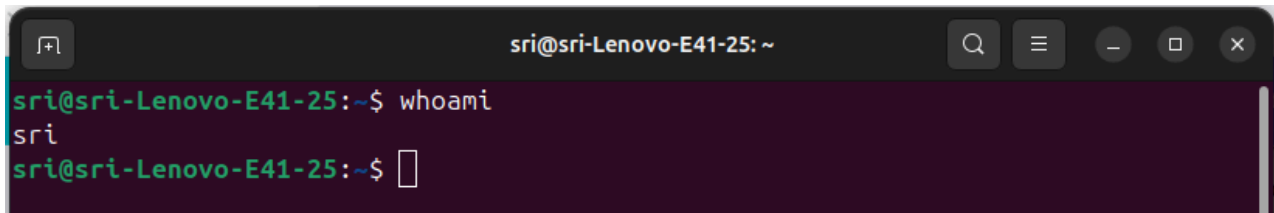
Category	Examples	Typical Extensions	Use Case
Text/Scripts	Plain text, Bash, Config	.txt, .sh, .conf, .cfg	Scripts, logs, configs
Binaries/Executables	ELF, Shared objects	(none), .bin, .so, .o	Programs, libraries
Archives/Compressed	tar, gzip, zip, rpm	.tar, .gz, .tgz, .rpm, .deb	Packages, backups
Images	PNG, JPEG	.png, .jpg, .svg	Graphics
Documents	PDF, RTF	.pdf, .rtf	Reports
Source Code	C, Python, Perl	.c, .cpp, .py, .pl	Development

4.Linux File Systems :

Filesystem	Key Features	Best Use Cases	Limitations
ext4	Journaling, extents for large files, reliable metadata	General-purpose (desktops, servers, root /)	No built-in snapshots or RAID
XFS	High performance with large files, parallel I/O, quotas	Databases, media servers, high-throughput workloads	No shrinking support
Btrfs	Snapshots, subvolumes, RAID, compression, self-healing	Advanced storage, backups, dynamic volumes	Higher complexity, potential stability issues
ZFS	Checksums, deduplication, RAID-Z, snapshots	Enterprise NAS, data integrity-critical setups	High RAM needs, licensing conflicts with Linux kernel

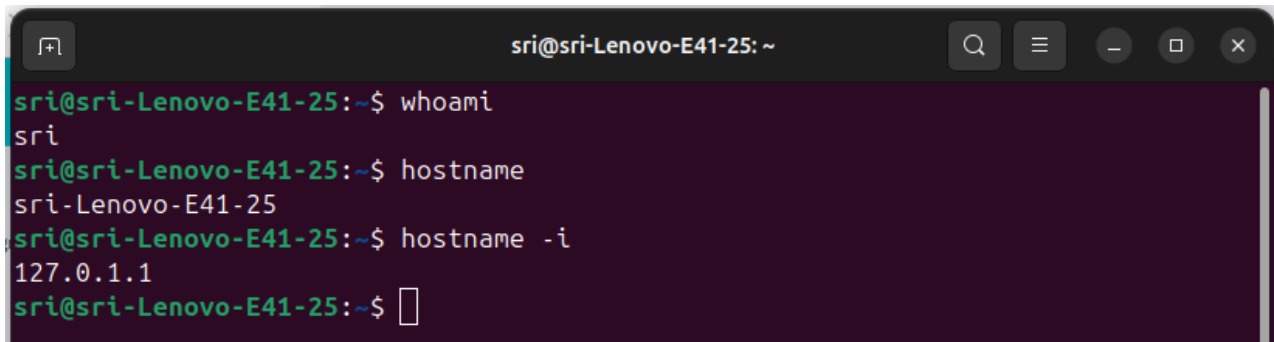
5.Linux Commands :

1.whoami - The username of the effective user

A terminal window titled 'sri@sri-Lenovo-E41-25: ~' showing the command 'whoami' being executed. The output is 'sri'.

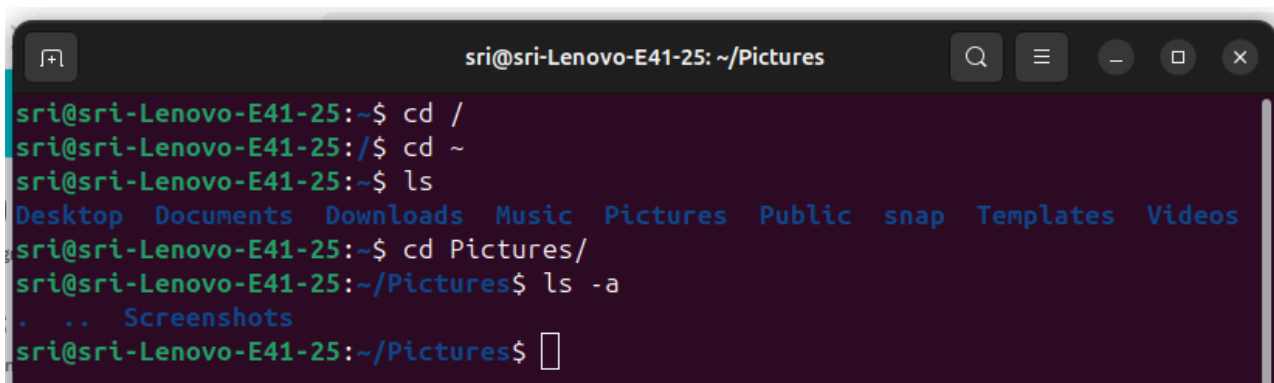
```
sri@sri-Lenovo-E41-25:~$ whoami
sri
sri@sri-Lenovo-E41-25:~$
```

2.Hostname-Show or set the system's network name (machine name).

A terminal window titled 'sri@sri-Lenovo-E41-25: ~' showing the commands 'hostname' and 'hostname -i'. The first command outputs 'sri-Lenovo-E41-25' and the second outputs '127.0.1.1'.

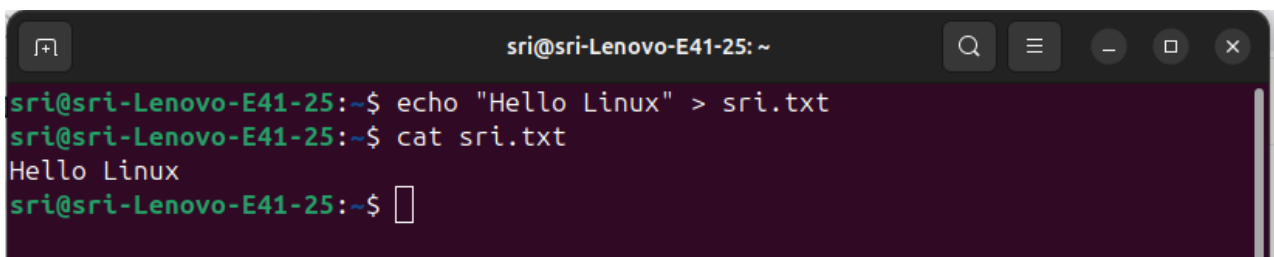
```
sri@sri-Lenovo-E41-25:~$ whoami
sri
sri@sri-Lenovo-E41-25:~$ hostname
sri-Lenovo-E41-25
sri@sri-Lenovo-E41-25:~$ hostname -i
127.0.1.1
sri@sri-Lenovo-E41-25:~$
```

3.cd – Change the current working directory.

A terminal window titled 'sri@sri-Lenovo-E41-25: ~/Pictures' showing a sequence of commands: 'cd /', 'cd ~', 'ls', 'cd Pictures/', and 'ls -a'. The output of 'ls' shows various system directories, and 'ls -a' in the Pictures directory shows '.', '..', and 'Screenshots'.

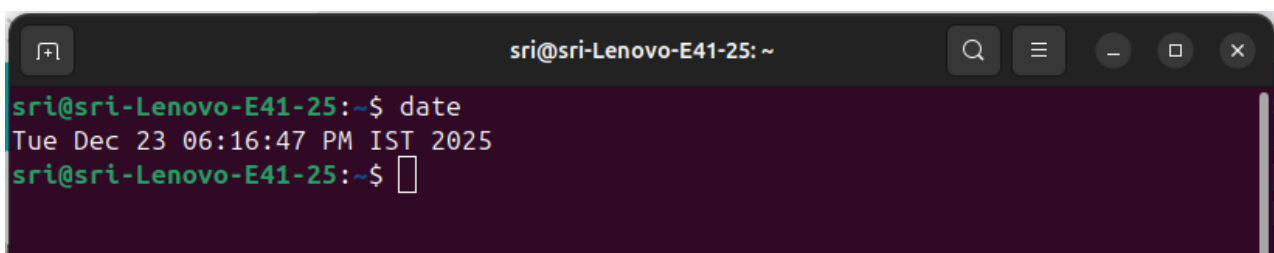
```
sri@sri-Lenovo-E41-25:~$ cd /
sri@sri-Lenovo-E41-25:/$ cd ~
sri@sri-Lenovo-E41-25:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
sri@sri-Lenovo-E41-25:~$ cd Pictures/
sri@sri-Lenovo-E41-25:~/Pictures$ ls -a
.  ..  Screenshots
sri@sri-Lenovo-E41-25:~/Pictures$
```

4.echo – Print text or variable values to the terminal

A terminal window titled 'sri@sri-Lenovo-E41-25: ~' showing the commands 'echo "Hello Linux" > sri.txt' and 'cat sri.txt'. The output of 'cat' is 'Hello Linux'.

```
sri@sri-Lenovo-E41-25:~$ echo "Hello Linux" > sri.txt
sri@sri-Lenovo-E41-25:~$ cat sri.txt
Hello Linux
sri@sri-Lenovo-E41-25:~$
```

5.date – Display or set the system date and time.

A terminal window titled 'sri@sri-Lenovo-E41-25: ~' showing the command 'date'. The output is 'Tue Dec 23 06:16:47 PM IST 2025'.

```
sri@sri-Lenovo-E41-25:~$ date
Tue Dec 23 06:16:47 PM IST 2025
sri@sri-Lenovo-E41-25:~$
```

6.sudo – Run a command with superuser (root) privileges.

```
root@sri-Lenovo-E41-25: /home/sri

sri@sri-Lenovo-E41-25:~$ sudo su
[sudo] password for sri:
root@sri-Lenovo-E41-25:/home/sri#
```

7.ls – List files and directories in a location.

```
sri@sri-Lenovo-E41-25: /

sri@sri-Lenovo-E41-25:/$ ls -a
.          cdrom  lib64      opt      sbin.usr-is-merged  tmp
..         dev   lib.usr-is-merged  proc     snap                usr
bin        etc   lost+found  root     srv                 var
bin.usr-is-merged  home  media      run      swap.img
boot       lib  mnt        sbin     sys
sri@sri-Lenovo-E41-25:/$
```

8.pwd – Show the full path of the current directory.

```
sri@sri-Lenovo-E41-25: ~/Pictures/Screenshots

sri@sri-Lenovo-E41-25:~/Pictures/Screenshots$ pwd
/home/sri/Pictures/Screenshots
sri@sri-Lenovo-E41-25:~/Pictures/Screenshots$
```

9.mkdir – Create a new directory.

```
sri@sri-Lenovo-E41-25: ~

sri@sri-Lenovo-E41-25:~$ mkdir sri
sri@sri-Lenovo-E41-25:~$ ls -a
.          .cache  .local  snap      Templates
..         .config Music    sri       Videos
.bash_history Desktop Pictures sri.txt
.bash_logout Documents .profile .ssh
.bashrc    Downloads Public   .sudo_as_admin_successful
sri@sri-Lenovo-E41-25:~$
```

10.rmdir – Remove an empty directory.

```
sri@sri-Lenovo-E41-25: ~

sri@sri-Lenovo-E41-25:~$ ls -a
.          .cache  .local  snap      Templates
..         .config Music    sri       Videos
.bash_history Desktop Pictures sri.txt
.bash_logout Documents .profile .ssh
.bashrc    Downloads Public   .sudo_as_admin_successful
sri@sri-Lenovo-E41-25:~$ rmdir sri/
sri@sri-Lenovo-E41-25:~$
```

11.touch – Create an empty file or update a file's timestamps.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ touch sri.txt  
sri@sri-Lenovo-E41-25:~$ nano sri.txt  
sri@sri-Lenovo-E41-25:~$ cat sri.txt  
Hello Sri!  
sri@sri-Lenovo-E41-25:~$
```

12.nano – Open a simple text editor in the terminal.

```
GNU nano 7.2 sri.txt  
Hello Sri!
```

13.cat (or) more – Display file contents or concatenate files.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ cat sri.txt  
Hello Sri!  
sri@sri-Lenovo-E41-25:~$
```

14.vi – Open the vi/vim text editor.

```
root@sri-Lenovo-E41-25: /home/sri  
GNU nano 7.2 /etc/sudoers.tmp  
#  
# This file MUST be edited with the 'visudo' command as root.  
#  
# Please consider adding local content in /etc/sudoers.d/ instead of  
# directly modifying this file.  
#  
# See the man page for details on how to write a sudoers file.  
#  
Defaults env_reset  
Defaults mail_badpass  
Defaults secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:  
# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu  
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)  
Defaults use_pty  
# This preserves proxy settings from user environments of root  
# equivalent users (group sudo)  
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_prox>  
[ Read 57 lines ]  
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location  
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line
```

15.mv – Move or rename files and directories.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ mv sri.txt soundar.txt  
sri@sri-Lenovo-E41-25:~$ ls -a  
.  
..  
.bash_history  
.bash_logout  
.bashrc  
.cache  
.config  
.desktop  
.local  
.profile  
.public  
.ssh  
.sudo_as_admin_successful  
Documents  
Downloads  
Music  
Pictures  
snap  
soundar.txt  
Templates  
Videos
```

16.cp – Copy files and directories.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ cp soundar.txt sri.txt  
sri@sri-Lenovo-E41-25:~$ ls -a  
.  
..  
.bash_history  
.bash_logout  
.bashrc  
.cache  
.config  
.desktop  
.local  
.profile  
.public  
.ssh  
.sudo_as_admin_successful  
Documents  
Downloads  
Music  
Pictures  
snap  
soundar.txt  
sri.txt  
Templates  
Videos
```

17.find – Search for files and directories by name or other criteria.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ find sri.txt  
sri.txt  
sri@sri-Lenovo-E41-25:~$
```

18.more – View file contents page by page (forward only).

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ more /var/log/syslog  
2025-12-19T06:33:26.367456+00:00 sri-Lenovo-E41-25 kernel: Linux version 6.14.0-  
27-generic (buildd@lcy02-amd64-120) (x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubu  
ntu2~24.04) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #27~24.04.1-Ubuntu SM  
P PREEMPT_DYNAMIC Tue Jul 22 17:38:49 UTC 2 (Ubuntu 6.14.0-27.27~24.04.1-generic  
6.14.6)  
2025-12-19T06:33:26.367704+00:00 sri-Lenovo-E41-25 kernel: Command line: BOOT_IM  
AGE=/boot/vmlinuz-6.14.0-27-generic root=UUID=70c37653-e255-4c9e-b0bf-2dbfe977d4  
c8 ro quiet splash vt.handoff=7  
2025-12-19T06:33:26.367710+00:00 sri-Lenovo-E41-25 kernel: KERNEL supported cpus  
:  
2025-12-19T06:33:26.367711+00:00 sri-Lenovo-E41-25 kernel: Intel GenuineIntel  
2025-12-19T06:33:26.367712+00:00 sri-Lenovo-E41-25 kernel: AMD AuthenticAMD  
2025-12-19T06:33:26.367713+00:00 sri-Lenovo-E41-25 kernel: Hygon HygonGenuine  
2025-12-19T06:33:26.367717+00:00 sri-Lenovo-E41-25 kernel: Centaur CentaurHaul  
s  
2025-12-19T06:33:26.367719+00:00 sri-Lenovo-E41-25 kernel: zhaoxin Shanghai  
2025-12-19T06:33:26.367721+00:00 sri-Lenovo-E41-25 kernel: BIOS-provided physica
```

19.ln – Create hard or symbolic links to files.

20.hard link – Additional directory entry pointing to the same node/data as an existing file.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ ln sri.txt sri_hard  
sri@sri-Lenovo-E41-25:~$ ls -a  
.          .cache      'Hi Sri !'  Public      .sudo_as_admin_successful  
..         .config     .local      snap        Templates  
.bash_history Desktop    Music       sri_hard    Videos  
.bash_logout Documents  Pictures     sri.txt  
.bashrc     Downloads .profile    .ssh  
sri@sri-Lenovo-E41-25:~$
```

20.chmod – Change file or directory permissions (read/write/execute).

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ touch sri.sh  
sri@sri-Lenovo-E41-25:~$ chmod 777 sri.sh  
sri@sri-Lenovo-E41-25:~$
```

21.soft link (symlink) – Special file that points to another path (like a shortcut).

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ ln -s /var/log/syslog syslog_link  
sri@sri-Lenovo-E41-25:~$ ls  
Desktop    Downloads  Pictures  snap        Templates  
Documents  Music      Public    syslog_link Videos  
sri@sri-Lenovo-E41-25:~$
```

22.gzip – Compress or decompress files using gzip format.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ gzip sri.txt  
sri@sri-Lenovo-E41-25:~$ ls -a  
.          .cache      .local      snap        Templates  
..         .config     Music       sri.txt.gz  Videos  
.bash_history Desktop    Pictures     .ssh  
.bash_logout Documents  .profile    .sudo_as_admin_successful  
.bashrc     Downloads Public      syslog_link  
sri@sri-Lenovo-E41-25:~$
```

23.uzip / unzip – Extract files from a ZIP archive.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ gzip sri.txt  
sri@sri-Lenovo-E41-25:~$ ls -a  
.          .cache      .local      snap        Templates  
..         .config     Music       sri.txt.gz  Videos  
.bash_history Desktop    Pictures     .ssh  
.bash_logout Documents  .profile    .sudo_as_admin_successful  
.bashrc     Downloads Public      syslog_link  
sri@sri-Lenovo-E41-25:~$ gunzip sri.txt.gz  
sri@sri-Lenovo-E41-25:~$
```


24.tar – Archive multiple files into one tarball (optionally compress).

```
sri@sri-Lenovo-E41-25:~$ tar -czf backup.tar.gz /home/user
tar: Removing leading '/' from member names
tar: /home/user: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
sri@sri-Lenovo-E41-25:~$ ls -la
.                  .bashrc           Downloads         Public            syslog_link
..                 .cache            .local           snap             Templates
backup.tar.gz      .config           Music             sri.txt          Videos
.bash_history      Desktop           Pictures          .ssh
.bash_logout       Documents         .profile         .sudo_as_admin_successful
sri@sri-Lenovo-E41-25:~$
```

25.alias – Define a custom shortcut name for a command.

```
sri@sri-Lenovo-E41-25:~$ alias ll='ls -la'

sri@sri-Lenovo-E41-25:~$ ll
total 80
drwxr-x--- 15 sri sri 4096 Dec 23 20:00 .
drwxr-xr-x  3 root root 4096 Dec 19 12:03 ..
-rw-rw-r--  1 sri sri   45 Dec 23 20:00 backup.tar.gz
-rw-----  1 sri sri  945 Dec 23 18:50 .bash_history
-rw-r--r--  1 sri sri  220 Mar 31 2024 .bash_logout
-rw-r--r--  1 sri sri 3771 Mar 31 2024 .bashrc
drwx----- 17 sri sri 4096 Dec 23 15:25 .cache
drwx----- 14 sri sri 4096 Dec 19 21:12 .config
drwxr-xr-x  2 sri sri 4096 Dec 19 12:04 Desktop
```

26.less – View file contents interactively, with forward/backward scrolling and search.

```
sri@sri-Lenovo-E41-25:~$ less /var/log/syslog
2025-12-19T06:33:26.367456+00:00 sri-Lenovo-E41-25 kernel: Linux version 6.14.0-27-generic (buildd@lcy02-amd64-120) (x86_64-linux-gnu-gcc-13 (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #27~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC Tue Jul 22 17:38:49 UTC 2 (Ubuntu 6.14.0-27.27~24.04.1-generic 6.14.6)
2025-12-19T06:33:26.367704+00:00 sri-Lenovo-E41-25 kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.14.0-27-generic root=UUID=70c37653-e255-4c9e-b0bf-2dbfe977d4c8 ro quiet splash vt.handoff=7
2025-12-19T06:33:26.367710+00:00 sri-Lenovo-E41-25 kernel: KERNEL supported cpus:
2025-12-19T06:33:26.367711+00:00 sri-Lenovo-E41-25 kernel: Intel GenuineIntel
2025-12-19T06:33:26.367712+00:00 sri-Lenovo-E41-25 kernel: AMD AuthenticAMD
2025-12-19T06:33:26.367713+00:00 sri-Lenovo-E41-25 kernel: Hygon HygonGenuine
2025-12-19T06:33:26.367717+00:00 sri-Lenovo-E41-25 kernel: Centaur CentaurHauls
2025-12-19T06:33:26.367719+00:00 sri-Lenovo-E41-25 kernel: zhaoxin Shanghai
2025-12-19T06:33:26.367721+00:00 sri-Lenovo-E41-25 kernel: BIOS-provided physical RAM map:
2025-12-19T06:33:26.367722+00:00 sri-Lenovo-E41-25 kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000000086fff] usable
2025-12-19T06:33:26.367723+00:00 sri-Lenovo-E41-25 kernel: BIOS-e820: [mem 0x000000000000087000-0x000000000000087fff] reserved
/var/log/syslog
```

27.tail – Show the last lines of a file (or follow new lines).

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ tail -f /var/log/syslog  
2025-12-23T20:11:25.129175+05:30 sri-Lenovo-E41-25 brave_brave.desktop[2953]: [3  
028:3033:1223/201125.126771:ERROR:services/network/restricted_cookie_manager.cc:  
1149] site_for_cookies from renderer='SiteForCookies: {site=null; schemefully_sa  
me=false}' from browser='SiteForCookies: {site=https://perplexity.ai; schemefull  
y_same=true}';  
2025-12-23T20:11:25.130794+05:30 sri-Lenovo-E41-25 brave_brave.desktop[2953]: [3  
028:3033:1223/201125.126835:ERROR:services/network/restricted_cookie_manager.cc:  
1158] top_frame_origin from renderer='null [internally: (3D443DA313F6422E049DA35  
48619F8D3) anonymous]' from browser='https://www.perplexity.ai';  
2025-12-23T20:11:56.056837+05:30 sri-Lenovo-E41-25 brave_brave.desktop[2953]: [3  
028:3033:1223/201156.054181:ERROR:services/network/restricted_cookie_manager.cc:  
1149] site_for_cookies from renderer='SiteForCookies: {site=null; schemefully_sa  
me=false}' from browser='SiteForCookies: {site=https://perplexity.ai; schemefull  
y_same=true}';  
2025-12-23T20:11:56.057458+05:30 sri-Lenovo-E41-25 brave_brave.desktop[2953]: [3  
028:3033:1223/201156.054223:ERROR:services/network/restricted_cookie_manager.cc:  
1158] top_frame_origin from renderer='null [internally: (A8C335BDA26ACAB93444847  
9F6DF75AE) anonymous]' from browser='https://www.perplexity.ai';  
2025-12-23T20:12:27.889163+05:30 sri-Lenovo-E41-25 brave_brave.desktop[2953]: [3  
028:3033:1223/201227.887459:ERROR:services/network/restricted_cookie_manager.cc:  
1149] site_for_cookies from renderer='SiteForCookies: {site=null; schemefully_sa  
me=false}' from browser='SiteForCookies: {site=https://perplexity.ai; schemefull
```

28.wc – Count lines, words, and bytes in files.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ nano sri.txt  
sri@sri-Lenovo-E41-25:~$ wc sri.txt  
 3  5 25 sri.txt  
sri@sri-Lenovo-E41-25:~$
```

29.grep – Search text for lines matching a pattern.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ grep "error" /var/log/syslog  
2025-12-19T06:33:26.374339+00:00 sri-Lenovo-E41-25 kernel: tpm_crb MSFT0101:00:  
error -EBUSY: can't request region for resource [mem 0xe7bb6000-0xe7bb9fff]  
2025-12-19T06:33:26.374340+00:00 sri-Lenovo-E41-25 kernel: tpm_crb MSFT0101:00:  
probe with driver tpm_crb failed with error -16  
2025-12-19T06:33:26.384218+00:00 sri-Lenovo-E41-25 systemd[1]: appport-autoreport  
.path - Process error reports when automatic reporting is enabled (file watch) w  
as skipped because of an unmet condition check (ConditionPathExists=/var/lib/app  
port/autoreport).  
2025-12-19T06:33:26.384253+00:00 sri-Lenovo-E41-25 systemd[1]: appport-autoreport  
.timer - Process error reports when automatic reporting is enabled (timer based)  
was skipped because of an unmet condition check (ConditionPathExists=/var/lib/a  
ppport/autoreport).  
2025-12-19T06:34:54.340901+00:00 sri-Lenovo-E41-25 gnome-initial-s[3553]: Failed  
to set text 'You can always enable Ubuntu Pro later via the Software & Updates
```

30.useradd (or) adduser - to create user in /home directory.

```
sri@sri-Lenovo-E41-25: /etc
sri@sri-Lenovo-E41-25:/$ sudo useradd -m -s /bin/bash raj
[sudo] password for sri:
sri@sri-Lenovo-E41-25:/$ cd /etc
sri@sri-Lenovo-E41-25:/etc$ tail passwd
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534:/:/run/gnome-initial-setup:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
sri:x:1000:1000:sri:/home/sri:/bin/bash
raj:x:1001:1001:/:/home/raj:/bin/bash
sri@sri-Lenovo-E41-25:/etc$
```

31.userdel - Delete a user (keep their home directory and files)

```
sri@sri-Lenovo-E41-25: /etc
sri@sri-Lenovo-E41-25:/etc$ sudo userdel raj
sri@sri-Lenovo-E41-25:/etc$ tail raj
tail: cannot open 'raj' for reading: No such file or directory
sri@sri-Lenovo-E41-25:/etc$ tail passwd
cups-browsed:x:115:114:/:/nonexistent:/usr/sbin/nologin
hplip:x:116:7:HPLIP system user,,,:/run/hplip:/bin/false
gnome-remote-desktop:x:988:988:GNOME Remote Desktop:/var/lib/gnome-remote-desktop:/usr/sbin/nologin
polkitd:x:987:987:User for polkitd:/:/usr/sbin/nologin
rtkit:x:117:119:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:118:120:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
gnome-initial-setup:x:119:65534:/:/run/gnome-initial-setup:/bin/false
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
sri:x:1000:1000:sri:/home/sri:/bin/bash
sri@sri-Lenovo-E41-25:/etc$
```

32.chown – Change file or directory owner and group.

```
root@sri-Lenovo-E41-25: ~  
gdm:x:120:121:Gnome Display Manager:/var/lib/gdm3:/bin/false  
nm-openvpn:x:121:122:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/n  
login  
sri:x:1000:1000:sri:/home/sri:/bin/bash  
raj:x:1001:1001::/home/raj:/bin/bash  
root@sri-Lenovo-E41-25:/etc# cd ~  
root@sri-Lenovo-E41-25:~# ls -a  
. .bash_history .cache .launchpadlib .profile sri.txt  
.. .bashrc .config .local snap .ssh  
root@sri-Lenovo-E41-25:~# ls -all  
total 44  
drwx----- 8 root root 4096 Dec 23 21:09 .  
drwxr-xr-x 23 root root 4096 Dec 19 11:58 ..  
-rw----- 1 root root 855 Dec 23 20:33 .bash_history  
-rw-r--r-- 1 root root 3106 Apr 22 2024 .bashrc  
drwx----- 6 root root 4096 Dec 23 15:36 .cache  
drwx----- 6 root root 4096 Dec 23 15:37 .config  
drwx----- 3 root root 4096 Dec 19 06:49 .launchpadlib  
drwxr-xr-x 3 root root 4096 Dec 19 12:09 .local  
-rw-r--r-- 1 root root 161 Apr 22 2024 .profile  
drwx----- 9 root root 4096 Dec 23 18:33 snap  
-rw-r--r-- 1 raj root 0 Dec 23 21:09 sri.txt  
drwx----- 2 root root 4096 Dec 19 12:03 .ssh  
root@sri-Lenovo-E41-25:~#
```

33.groupadd - to create a group on /root directory.

```
root@sri-Lenovo-E41-25: /etc  
root@sri-Lenovo-E41-25:/# groupadd Devops  
root@sri-Lenovo-E41-25:/# cd /etc  
root@sri-Lenovo-E41-25:/etc# tail group  
rtkit:x:119:  
colord:x:120:  
gdm:x:121:  
nm-openvpn:x:122:  
lxd:x:123:  
gamemode:x:986:  
gnome-initial-setup:x:985:  
sri:x:1000:  
raj:x:1001:  
Devops:x:1002:  
root@sri-Lenovo-E41-25:/etc#
```


34.usermod - Add existing user to the group.

```
root@sri-Lenovo-E41-25: /etc
root@sri-Lenovo-E41-25:/# sudo usermod -aG Devops raj
root@sri-Lenovo-E41-25:/# cd /etc
root@sri-Lenovo-E41-25:/etc# tail group
rtkit:x:119:
colord:x:120:
gdm:x:121:
nm-openvpn:x:122:
lxd:x:123:
gamemode:x:986:
gnome-initial-setup:x:985:
sri:x:1000:
raj:x:1001:
Devops:x:1002:raj
root@sri-Lenovo-E41-25:/etc#
```

33.chgrp – Change the group ownership of a file or directory.

```
root@sri-Lenovo-E41-25: /
drwxr-xr-x  4 root root      4096 Dec 19 21:20 boot
drwxr-xr-x  2 root root      4096 Jan  1 1970 cdrom
drwxr-xr-x 20 root root      4520 Dec 23 18:33 dev
drwxr-xr-x 139 root root    12288 Dec 23 21:47 etc
drwxr-xr-x  4 root root      4096 Dec 23 20:58 home
lrwxrwxrwx  1 root root         7 Apr 22 2024 lib -> usr/lib
lrwxrwxrwx  1 root root         9 Apr 22 2024 lib64 -> usr/lib64
drwxr-xr-x  2 root root      4096 Apr  8 2024 lib.usr-is-merged
drwx----- 2 root root    16384 Dec 19 11:50 lost+found
drwxr-xr-x  2 root root      4096 Aug  5 22:18 media
drwxr-xr-x  2 root root      4096 Aug  5 22:18 mnt
drwxr-xr-x  2 root root      4096 Aug  5 22:18 opt
dr-xr-xr-x 316 root root         0 Dec 23 18:01 proc
drwx----- 8 root root      4096 Dec 23 21:09 root
drwxr-xr-x 37 root root       900 Dec 23 18:19 run
lrwxrwxrwx  1 root root         8 Apr 22 2024 sbin -> usr/sbin
drwxr-xr-x  2 root root      4096 Mar 31 2024 sbin.usr-is-merged
drwxr-xr-x 17 root root      4096 Dec 23 18:33 snap
-rw-r--r--  1 root Devops         0 Dec 23 21:49 sri.txt
drwxr-xr-x  2 root root      4096 Aug  5 22:18 srv
-rw-----  1 root root    3939500032 Dec 19 11:58 swap.img
dr-xr-xr-x 13 root root         0 Dec 23 18:23 sys
drwxrwxrwt 19 root root      4096 Dec 23 21:47 tmp
drwxr-xr-x 12 root root      4096 Aug  5 22:18 usr
drwxr-xr-x 14 root root      4096 Dec 19 12:03 var
```

34.shutdown – Power off or reboot the system at once or at a scheduled time.

sudo shutdown +5

- power off in 5 minutes

35.tree – Display directories and files in a tree-like structure.

```
root@sri-Lenovo-E41-25: /  
root@sri-Lenovo-E41-25:/# tree /etc  
/etc  
├── adduser.conf  
├── alsa  
│   └── conf.d  
│       ├── 50-pipewire.conf -> /usr/share/alsa/alsa.conf.d/50-pipewire.conf  
│       └── 99-pipewire-default.conf -> /usr/share/alsa/alsa.conf.d/99-pipewire-de  
fault.conf  
├── alternatives  
│   ├── arptables -> /usr/sbin/arptables-nft  
│   ├── arptables-restore -> /usr/sbin/arptables-nft-restore  
│   ├── arptables-save -> /usr/sbin/arptables-nft-save  
│   ├── awk -> /usr/bin/mawk  
│   ├── awk.1.gz -> /usr/share/man/man1/mawk.1.gz  
│   ├── builtins.7.gz -> /usr/share/man/man7/bash-builtins.7.gz  
│   ├── cpp -> /usr/bin/cpp  
│   ├── default.plymouth -> /usr/share/plymouth/themes/bgrt/bgrt.plymouth  
│   ├── ebtables -> /usr/sbin/ebtables-nft  
│   ├── ebtables-restore -> /usr/sbin/ebtables-nft-restore  
│   ├── ebtables-save -> /usr/sbin/ebtables-nft-save  
│   ├── editor -> /bin/nano  
│   ├── editor.1.gz -> /usr/share/man/man1/nano.1.gz  
│   ├── ex -> /usr/bin/vim.tiny  
│   └── ex.1.gz -> /usr/share/man/man1/vim.1.gz
```

36.uname -a – system/kernel info.

```
root@sri-Lenovo-E41-25: /  
root@sri-Lenovo-E41-25:/# uname -a  
Linux sri-Lenovo-E41-25 6.14.0-37-generic #37~24.04.1-Ubuntu SMP PREEMPT_DYNAMIC T  
hu Nov 20 10:25:38 UTC 2 x86_64 x86_64 x86_64 GNU/Linux  
root@sri-Lenovo-E41-25:/#
```

37.head – first 10 lines.

```
root@sri-Lenovo-E41-25: /etc  
root@sri-Lenovo-E41-25:/etc# head passwd  
root:x:0:0:root:/root:/bin/bash  
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin  
bin:x:2:2:bin:/bin:/usr/sbin/nologin  
sys:x:3:3:sys:/dev:/usr/sbin/nologin  
sync:x:4:65534:sync:/bin:/bin/sync  
games:x:5:60:games:/usr/games:/usr/sbin/nologin  
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin  
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin  
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin  
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin  
root@sri-Lenovo-E41-25:/etc#
```

38.ps aux – list processes.

```
root@sri-Lenovo-E41-25: /etc# ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.3 23612 14012 ?        Ss   18:01   0:13 /sbin/init spla
root           2  0.0  0.0      0     0 ?        S    18:01   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    18:01   0:00 [pool_workqueue
root           4  0.0  0.0      0     0 ?        I<   18:01   0:00 [kworker/R-rcu_
root           5  0.0  0.0      0     0 ?        I<   18:01   0:00 [kworker/R-sync
root           6  0.0  0.0      0     0 ?        I<   18:01   0:00 [kworker/R-kvfr
root           7  0.0  0.0      0     0 ?        I<   18:01   0:00 [kworker/R-slub
root           8  0.0  0.0      0     0 ?        I<   18:01   0:00 [kworker/R-netn
root          11  0.0  0.0      0     0 ?        I<   18:01   0:00 [kworker/0:0H-e
root          13  0.0  0.0      0     0 ?        I<   18:01   0:00 [kworker/R-mm_p
root          14  0.0  0.0      0     0 ?        I    18:01   0:00 [rcu_tasks_kthr
root          15  0.0  0.0      0     0 ?        I    18:01   0:00 [rcu_tasks_rude
root          16  0.0  0.0      0     0 ?        I    18:01   0:00 [rcu_tasks_trac
root          17  0.0  0.0      0     0 ?        S    18:01   0:04 [ksoftirqd/0]
root          18  0.1  0.0      0     0 ?        I    18:01   0:24 [rcu_preempt]
root          19  0.0  0.0      0     0 ?        S    18:01   0:00 [rcu_exp_par_gp
root          20  0.0  0.0      0     0 ?        S    18:01   0:00 [rcu_exp_gp_kth
root          21  0.0  0.0      0     0 ?        S    18:01   0:00 [migration/0]
root          22  0.0  0.0      0     0 ?        S    18:01   0:00 [idle_inject/0]
root          23  0.0  0.0      0     0 ?        S    18:01   0:00 [cpuhp/0]
root          24  0.0  0.0      0     0 ?        S    18:01   0:00 [cpuhp/1]
root          25  0.0  0.0      0     0 ?        S    18:01   0:00 [idle_inject/1]
```

39.top – live process view.

```
root@sri-Lenovo-E41-25: /
top - 22:12:33 up 4:11, 1 user, load average: 0.42, 0.82, 0.92
Tasks: 251 total, 1 running, 250 sleeping, 0 stopped, 0 zombie
%Cpu(s): 6.1 us, 2.2 sy, 0.0 ni, 90.4 id, 1.4 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3757.5 total, 129.9 free, 2777.2 used, 1120.9 buff/cache
MiB Swap: 3757.0 total, 2861.2 free, 895.8 used. 980.4 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
 2067 sri        20   0 3999252 177328 56964 S   7.9   4.6   12:43.15 gnome-shell
 8073 sri        20   0 624432   53444 41544 S   4.3   1.4    0:28.15 gnome-termi+
 3237 sri        20   0 1393.7g 601984 144904 S   2.0  15.6   27:06.94 brave
 2954 sri        20   0  48.9g 249884 161944 S   1.7   6.5    8:58.67 brave
11000 root        20   0  14536   5888   3680 R   0.7   0.2    0:00.10 top
  732 systemd+   20   0  17560   6648   6492 S   0.3   0.2    0:07.53 systemd-oomd
 3022 sri        20   0  48.7g 162456 121168 S   0.3   4.2   11:48.72 brave
10918 root       20   0      0      0      0 I   0.3   0.0    0:00.44 kworker/u17+
    1 root       20   0 23612   13564  9084 S   0.0   0.4    0:13.59 systemd
    2 root       20   0      0      0      0 S   0.0   0.0    0:00.02 kthreadd
    3 root       20   0      0      0      0 S   0.0   0.0    0:00.00 pool_workqu+
    4 root        0 -20      0      0      0 I   0.0   0.0    0:00.00 kworker/R-r+
    5 root        0 -20      0      0      0 I   0.0   0.0    0:00.00 kworker/R-s+
    6 root        0 -20      0      0      0 I   0.0   0.0    0:00.00 kworker/R-k+
    7 root        0 -20      0      0      0 I   0.0   0.0    0:00.00 kworker/R-s+
    8 root        0 -20      0      0      0 I   0.0   0.0    0:00.00 kworker/R-n+
   11 root        0 -20      0      0      0 I   0.0   0.0    0:00.00 kworker/0:0+
```

40.history – show past commands.

```
sri@sri-Lenovo-E41-25: ~  
sri@sri-Lenovo-E41-25:~$ history | tail  
320 cd /etc  
321 tail paswd  
322 tail passwd  
323 clear  
324 cd /  
325 sudo su  
326 sudo systemctl stop apache2  
327 sudo systemctl status  
328 clear  
329 history | tail  
sri@sri-Lenovo-E41-25:~$
```

41.wget URL – download file

```
root@sri-Lenovo-E41-25: /  
root@sri-Lenovo-E41-25:/# wget https://releases.ubuntu.com/24.04/ubuntu-24.04-de  
sktop-amd64.iso  
--2025-12-24 13:05:36-- https://releases.ubuntu.com/24.04/ubuntu-24.04-desktop-  
amd64.iso  
Resolving releases.ubuntu.com (releases.ubuntu.com)... 2620:2d:4000:1::1a, 2620:  
2d:4002:1::108, 2620:2d:4002:1::109, ...  
Connecting to releases.ubuntu.com (releases.ubuntu.com)|2620:2d:4000:1::1a|:443.  
.. connected.
```

42.kill – stop process.

```
root@sri-Lenovo-E41-25: /var/run/apache2  
root@sri-Lenovo-E41-25:/# cd /var/run/apache2/  
root@sri-Lenovo-E41-25:/var/run/apache2# more apache2.pid  
6115  
root@sri-Lenovo-E41-25:/var/run/apache2# kill 6115  
root@sri-Lenovo-E41-25:/var/run/apache2# systemctl status apache2  
○ apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >  
   Active: inactive (dead) since Wed 2025-12-24 13:18:27 IST; 33s ago  
   Duration: 9min 27.665s  
   Docs: https://httpd.apache.org/docs/2.4/  
   Process: 6112 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU>  
   Process: 6304 ExecStop=/usr/sbin/apachectl graceful-stop (code=exited, stat>  
   Main PID: 6115 (code=exited, status=0/SUCCESS)  
   CPU: 174ms  
  
Dec 24 13:08:59 sri-Lenovo-E41-25 systemd[1]: Starting apache2.service - The Ap>  
Dec 24 13:08:59 sri-Lenovo-E41-25 apachectl[6114]: AH00558: apache2: Could not >  
Dec 24 13:08:59 sri-Lenovo-E41-25 systemd[1]: Started apache2.service - The Apa>  
Dec 24 13:18:27 sri-Lenovo-E41-25 apachectl[6306]: AH00558: apache2: Could not >  
Dec 24 13:18:27 sri-Lenovo-E41-25 apachectl[6306]: httpd (no pid file) not runn>  
Dec 24 13:18:27 sri-Lenovo-E41-25 systemd[1]: apache2.service: Deactivated succ>
```


43.man cmd – manual page.

```
root@sri-Lenovo-E41-25: /
LS(1) User Commands LS(1)

NAME
    ls - list directory contents

SYNOPSIS
    ls [OPTION]... [FILE]...

DESCRIPTION
    List information about the FILES (the current directory by default).
    Sort entries alphabetically if none of -cftuvSUX nor --sort is speci-
    fied.

    Mandatory arguments to long options are mandatory for short options
    too.

    -a, --all
        do not ignore entries starting with .

    -A, --almost-all
        do not list implied . and ..

    --author
    Manual page ls(1) line 1 (press h for help or q to quit)
```

44.cmd --help – quick options help.

```
root@sri-Lenovo-E41-25: /
root@sri-Lenovo-E41-25: /# grep --help
Usage: grep [OPTION]... PATTERNS [FILE]...
Search for PATTERNS in each FILE.
Example: grep -i 'hello world' menu.h main.c
PATTERNS can contain multiple patterns separated by newlines.

Pattern selection and interpretation:
  -E, --extended-regexp  PATTERNS are extended regular expressions
  -F, --fixed-strings     PATTERNS are strings
  -G, --basic-regexp     PATTERNS are basic regular expressions
  -P, --perl-regexp      PATTERNS are Perl regular expressions
  -e, --regexp=PATTERNS  use PATTERNS for matching
  -f, --file=FILE        take PATTERNS from FILE
  -i, --ignore-case       ignore case distinctions in patterns and data
                        --no-ignore-case do not ignore case distinctions (default)
  -w, --word-regexp      match only whole words
  -x, --line-regexp      match only whole lines
  -z, --null-data        a data line ends in 0 byte, not newline

Miscellaneous:
  -s, --no-messages      suppress error messages
  -v, --invert-match      select non-matching lines
  -V, --version           display version information and exit
  --help                 display this help text and exit
```

45.htop – interactive process viewer.

```
root@sri-Lenovo-E41-25: /

0[|||||] 19.4% Tasks: 140, 581 thr, 116 kthr; 1 runni
1[||||] 9.3% Load average: 0.72 0.97 0.97
Mem[|||||||||||||||||] 2.55G/3.67G Uptime: 03:28:06
Swp[|||] 214M/3.67G

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
2268 sri 20 0 3902M 208M 61164 S 15.9 5.6 8:05.14 /usr/bin/gnom
6736 root 20 0 11920 5548 3612 R 5.7 0.1 0:00.62 htop
2292 sri -21 0 3902M 208M 61164 S 1.9 5.6 1:37.51 /usr/bin/gnom
2316 sri 20 0 3902M 208M 61164 S 1.3 5.6 1:05.58 /usr/bin/gnom
3279 sri 20 0 48.8G 305M 217M S 0.6 8.1 1:10.33 /snap/brave/5
3280 sri 20 0 48.8G 305M 217M S 0.6 8.1 0:34.56 /snap/brave/5
3715 sri 20 0 615M 49636 37316 S 0.6 1.3 0:08.70 /usr/libexec/
1 root 20 0 23660 14936 9480 S 0.0 0.4 0:06.49 /sbin/init sp
264 root 19 -1 67240 17640 16980 S 0.0 0.5 0:01.16 /usr/lib/syst
352 root 20 0 30716 8460 4656 S 0.0 0.2 0:00.43 /usr/lib/syst
839 systemd-oo 20 0 17560 7572 6656 S 0.0 0.2 0:06.25 /usr/lib/syst
850 systemd-re 20 0 21824 12828 10304 S 0.0 0.3 0:03.29 /usr/lib/syst
855 systemd-ti 20 0 91048 7144 6908 S 0.0 0.2 0:00.22 /usr/lib/syst
885 systemd-ti 20 0 91048 7144 6908 S 0.0 0.2 0:00.00 /usr/lib/syst
944 avahi 20 0 8672 4392 3936 S 0.0 0.1 0:00.22 avahi-daemon:
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice -F8Nice +F9Kill F10Quit
```

46.strace – trace system calls of a process

```
root@sri-Lenovo-E41-25: /

root@sri-Lenovo-E41-25: /# systemctl start apache2
root@sri-Lenovo-E41-25: /# cd var/run/apache2/
root@sri-Lenovo-E41-25: /var/run/apache2# cat apache2.pid
6803
root@sri-Lenovo-E41-25: /var/run/apache2# cd /
root@sri-Lenovo-E41-25: /# strace -p 6803
strace: Process 6803 attached
pselect6(0, NULL, NULL, NULL, {tv_sec=0, tv_nsec=787109952}, NULL) = 0 (Timeout)
wait4(-1, 0x7ffd1b234124, WNOHANG|WSTOPPED, NULL) = 0
times({tms_utime=1 /* 0.01 s */, tms_stime=1 /* 0.01 s */, tms_cutime=0, tms_cstime=0}) = 430747847
pselect6(0, NULL, NULL, NULL, {tv_sec=1, tv_nsec=0}, NULL) = 0 (Timeout)
wait4(-1, 0x7ffd1b234124, WNOHANG|WSTOPPED, NULL) = 0
times({tms_utime=1 /* 0.01 s */, tms_stime=1 /* 0.01 s */, tms_cutime=0, tms_cstime=0}) = 430747947
pselect6(0, NULL, NULL, NULL, {tv_sec=1, tv_nsec=0}, NULL) = 0 (Timeout)
wait4(-1, 0x7ffd1b234124, WNOHANG|WSTOPPED, NULL) = 0
times({tms_utime=1 /* 0.01 s */, tms_stime=1 /* 0.01 s */, tms_cutime=0, tms_cstime=0}) = 430748047
pselect6(0, NULL, NULL, NULL, {tv_sec=1, tv_nsec=0}, NULL) = 0 (Timeout)
wait4(-1, 0x7ffd1b234124, WNOHANG|WSTOPPED, NULL) = 0
times({tms_utime=1 /* 0.01 s */, tms_stime=1 /* 0.01 s */, tms_cutime=0, tms_cstime=0}) = 430748148
pselect6(0, NULL, NULL, NULL, {tv_sec=1, tv_nsec=0}, NULL) = 0 (Timeout)
```

47.lsof – list open files / ports.

```

root@sri-Lenovo-E41-25: /
8,6      43376    2651639 /usr/lib/apache2/modules/mod_autoindex.so
apache2   6806    6835    apache2          www-data  mem      REG
8,6      14704    2651638 /usr/lib/apache2/modules/mod_authz_user.so
apache2   6806    6835    apache2          www-data  mem      REG
8,6      14704    2651636 /usr/lib/apache2/modules/mod_authz_host.so
apache2   6806    6835    apache2          www-data  mem      REG
8,6      31088    2651632 /usr/lib/apache2/modules/mod_authz_core.so
apache2   6806    6835    apache2          www-data  mem      REG
8,6      18800    2651621 /usr/lib/apache2/modules/mod_auth_basic.so
apache2   6806    6835    apache2          www-data  mem      REG
8,6      22896    2651618 /usr/lib/apache2/modules/mod_alias.so
apache2   6806    6835    apache2          www-data  mem      REG
8,6      14704    2651628 /usr/lib/apache2/modules/mod_authn_file.so
apache2   6806    6835    apache2          www-data  mem      REG
8,6      14704    2651625 /usr/lib/apache2/modules/mod_authn_core.so
apache2   6806    6835    apache2          www-data  mem      REG
8,6      2125328    2117159 /usr/lib/x86_64-linux-gnu/libc.so.6
apache2   6806    6835    apache2          www-data  mem      REG
8,6      14704    2651616 /usr/lib/apache2/modules/mod_access_compat.so
apache2   6806    6835    apache2          www-data  mem      REG
8,6      35032    2101856 /usr/lib/x86_64-linux-gnu/libuuid.so.1.3.0
apache2   6806    6835    apache2          www-data  mem      REG
8,6      198664    2112770 /usr/lib/x86_64-linux-gnu/libcrypt.so.1.1.0
apache2   6806    6835    apache2          www-data  mem      REG

```

48.lsbblk – show block devices and mounts

```

root@sri-Lenovo-E41-25: /
root@sri-Lenovo-E41-25: /# lsblk -f
NAME FSTYPE FSVER LABEL UUID FSAVAIL FSUSE%
MOUNTPOINTS
loop0
  squash 4.0 0 100%
  /snap/bare/5
loop1
  squash 4.0 0 100%
  /snap/core24/1237
loop2
  squash 4.0 0 100%
  /snap/core22/2163
loop3
  squash 4.0 0 100%
  /snap/gnome-42-2204/202
loop4
  squash 4.0 0 100%
  /snap/firefox/7559
loop5
  squash 4.0 0 100%
  /snap/firmware-updater/210
loop6
  squash 4.0 0 100%
  /snap/firefox/6565

```

49.df -h – disk space summary

```
root@sri-Lenovo-E41-25: /  
root@sri-Lenovo-E41-25:/# df -h  
Filesystem      Size  Used Avail Use% Mounted on  
tmpfs           376M  2.0M  374M   1% /run  
/dev/sda6       59G   14G   43G  24% /  
tmpfs           1.9G   82M   1.8G   5% /dev/shm  
tmpfs           5.0M   12K   5.0M   1% /run/lock  
efivarfs        84K   32K   48K  40% /sys/firmware/efi/efivars  
/dev/sda1       96M   32M   65M  33% /boot/efi  
tmpfs           376M  128K  376M   1% /run/user/1000  
root@sri-Lenovo-E41-25:/#
```

50.du -sh – size of a directory

```
root@sri-Lenovo-E41-25: /  
root@sri-Lenovo-E41-25:/# du -sh /var/log  
139M    /var/log  
root@sri-Lenovo-E41-25:/#
```

51.ip a – show interfaces and IPs

```
root@sri-Lenovo-E41-25: /  
root@sri-Lenovo-E41-25:/# ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default  
qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp1s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN  
group default qlen 1000  
    link/ether 9c:5a:44:5d:a5:7d brd ff:ff:ff:ff:ff:ff  
3: wlp2s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group  
default qlen 1000  
    link/ether 3c:91:80:60:9d:8d brd ff:ff:ff:ff:ff:ff  
    inet 10.37.244.215/24 brd 10.37.244.255 scope global dynamic noprefixroute wlp  
2s0  
        valid_lft 2645sec preferred_lft 2645sec  
    inet6 2401:4900:9271:5eaf:b68a:1ba3:8635:1e3d/64 scope global temporary dynami  
c  
        valid_lft 6697sec preferred_lft 6697sec  
    inet6 2401:4900:9271:5eaf:29fb:2212:9f94:384f/64 scope global dynamic mngtmpad  
dr noprefixroute  
        valid_lft 6697sec preferred_lft 6697sec  
    inet6 fe80::2d8d:e93c:9053:dcba/64 scope link noprefixroute
```

52.ss -nltu – show listening ports

```
root@sri-Lenovo-E41-25: /  
root@sri-Lenovo-E41-25:/# sudo ss -nltu  
Netid  State  Recv-Q  Send-Q  Local Address:Port  Peer Address:Port  Process  
udp    UNCONN 0        0        127.0.0.54:53       0.0.0.0:*  
udp    UNCONN 0        0        127.0.0.53%lo:53    0.0.0.0:*  
udp    UNCONN 0        0        0.0.0.0:58580       0.0.0.0:*  
udp    UNCONN 0        0        224.0.0.251:5353    0.0.0.0:*  
udp    UNCONN 0        0        0.0.0.0:5353        0.0.0.0:*  
udp    UNCONN 0        0        [::]:59069          [::]:*  
udp    UNCONN 0        0        [::]:5353           [::]:*  
tcp    LISTEN 0        4096     127.0.0.1:631        0.0.0.0:*  
tcp    LISTEN 0        4096     127.0.0.54:53       0.0.0.0:*  
tcp    LISTEN 0        4096     0.0.0.0:22          0.0.0.0:*  
tcp    LISTEN 0        4096     127.0.0.53%lo:53    0.0.0.0:*  
tcp    LISTEN 0        4096     [::]:22             [::]:*  
tcp    LISTEN 0        511      *:80                 *:.*  
tcp    LISTEN 0        4096     [::1]:631           [::]:*
```

53.tcpdump – capture packets

```
root@sri-Lenovo-E41-25: /  
root@sri-Lenovo-E41-25:/# sudo tcpdump -i eth0 port 80  
tcpdump: eth0: No such device exists  
(No such device exists)  
root@sri-Lenovo-E41-25:/# sudo tcpdump -i eth0 port 443  
tcpdump: eth0: No such device exists  
(No such device exists)
```

54.awk – field-based text processing

```
root@sri-Lenovo-E41-25: /  
2025-12-24T14:00:44.566509+05:30 top_frame_origin  
root@sri-Lenovo-E41-25:/# awk '{print $1, $5}' /var/log/syslog | tail  
2025-12-24T14:01:23.506539+05:30 type=1400  
2025-12-24T14:01:23.509349+05:30 type=1400  
2025-12-24T14:01:23.512261+05:30 type=1400  
2025-12-24T14:01:23.512291+05:30 type=1400  
2025-12-24T14:01:23.513177+05:30 type=1400  
2025-12-24T14:01:23.515185+05:30 type=1400  
2025-12-24T14:01:23.516179+05:30 type=1400  
2025-12-24T14:01:23.518182+05:30 type=1400  
2025-12-24T14:01:23.519177+05:30 type=1400  
2025-12-24T14:01:23.519195+05:30 type=1400  
root@sri-Lenovo-E41-25:/# awk '{print $1, $5}' /var/log/syslog | head  
2025-12-19T06:33:26.367456+00:00 version  
2025-12-19T06:33:26.367704+00:00 line:  
2025-12-19T06:33:26.367710+00:00 supported  
2025-12-19T06:33:26.367711+00:00 GenuineIntel  
2025-12-19T06:33:26.367712+00:00 AuthenticAMD  
2025-12-19T06:33:26.367713+00:00 HygonGenuine  
2025-12-19T06:33:26.367717+00:00 CentaurHauls  
2025-12-19T06:33:26.367719+00:00 Shanghai  
2025-12-19T06:33:26.367721+00:00 physical  
2025-12-19T06:33:26.367722+00:00 [mem
```